Course end project

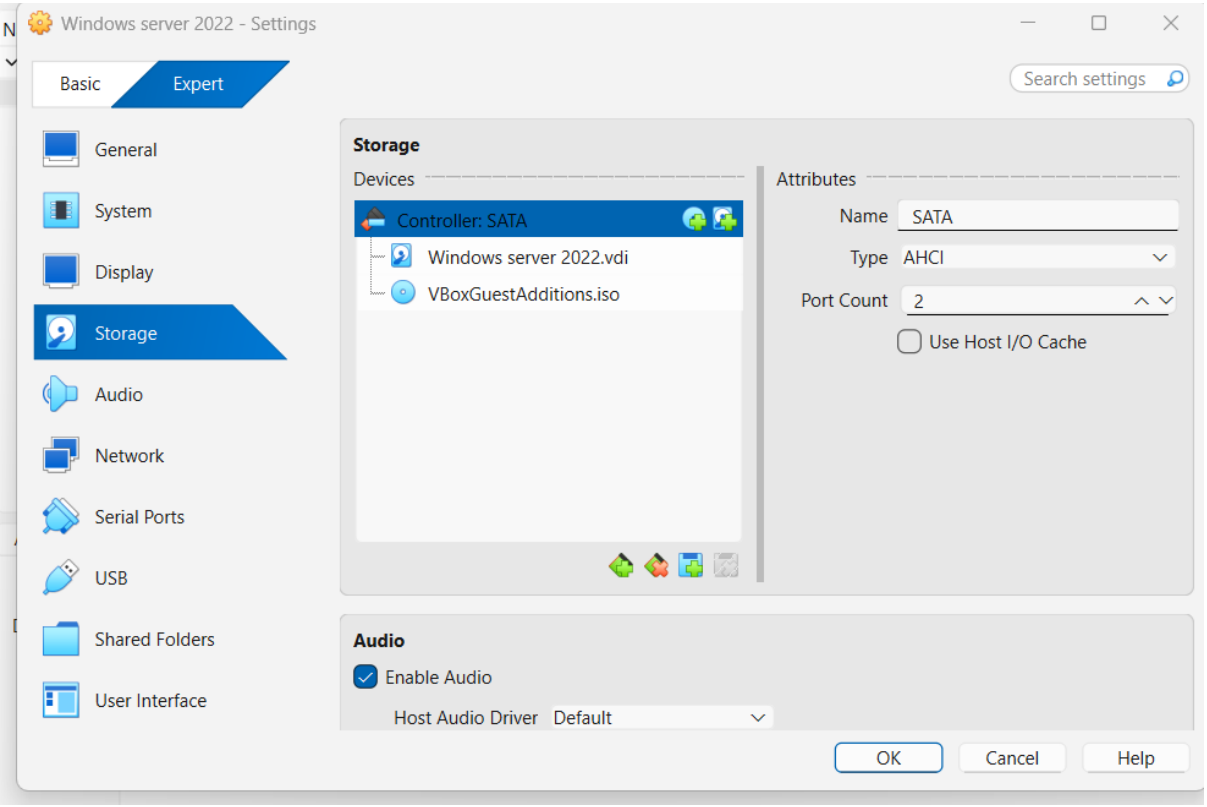
Recovering Hidden Data from Jpeg Image

Steps to be followed:

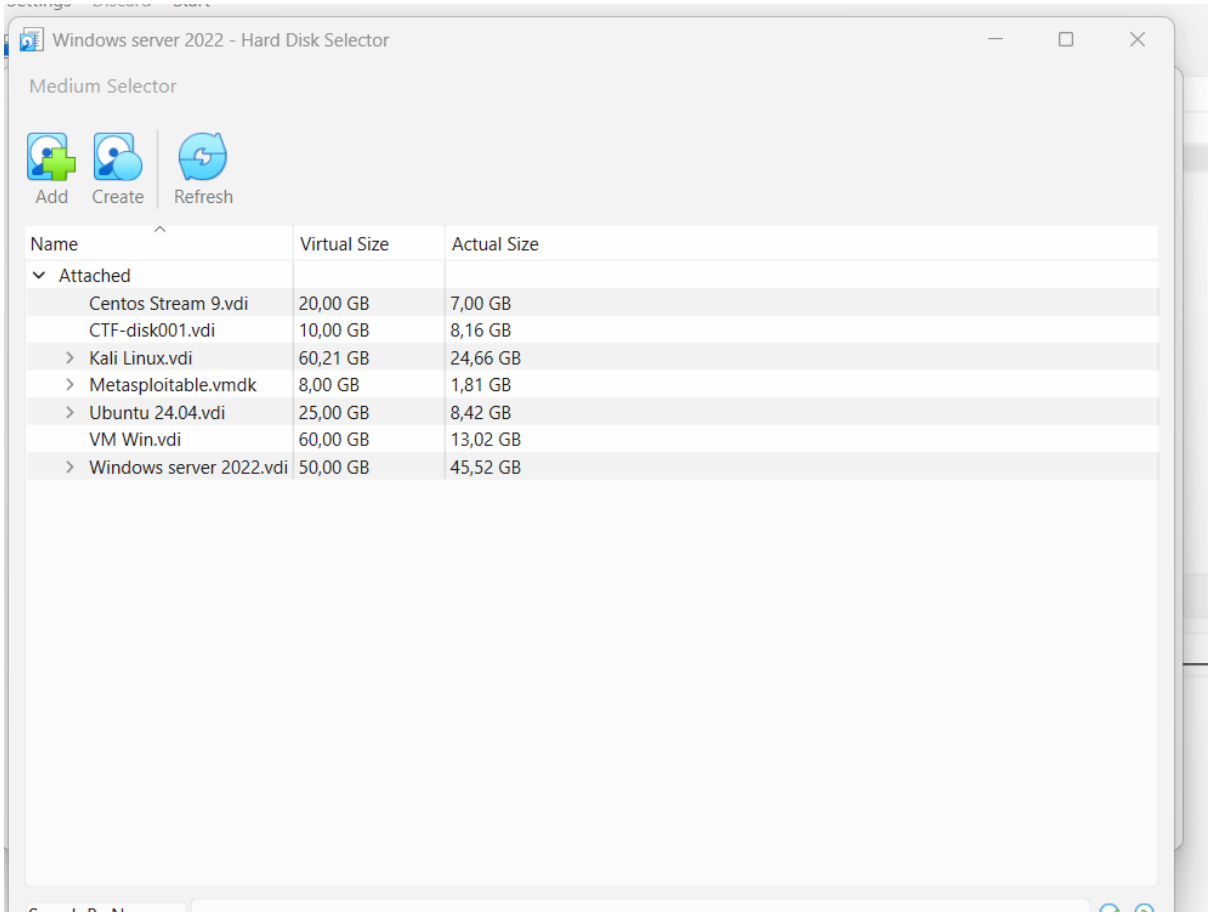
1. Create a JPG image and hide text using steganography
2. Download and prepare the data to be tested
3. Download Access data FTK Imager
4. Hide text in the image using Steganography
5. Store the image on a 2GB Disk in NTFS
6. Recover the deleted file using FTK

Step 1: Create a JPG image and hide text using steganography

1.1 Click windows server storage

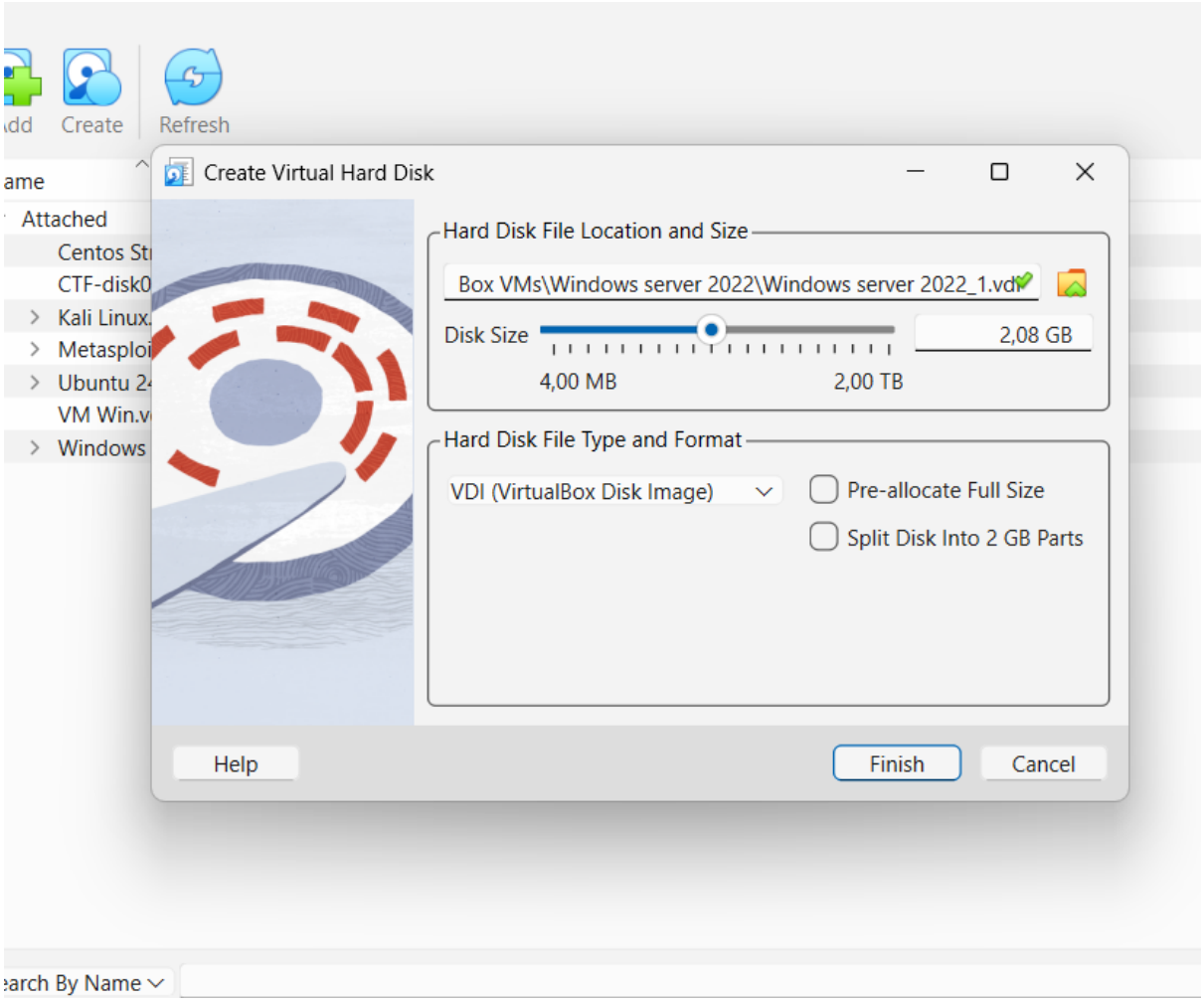


1.2 click Hard Disk

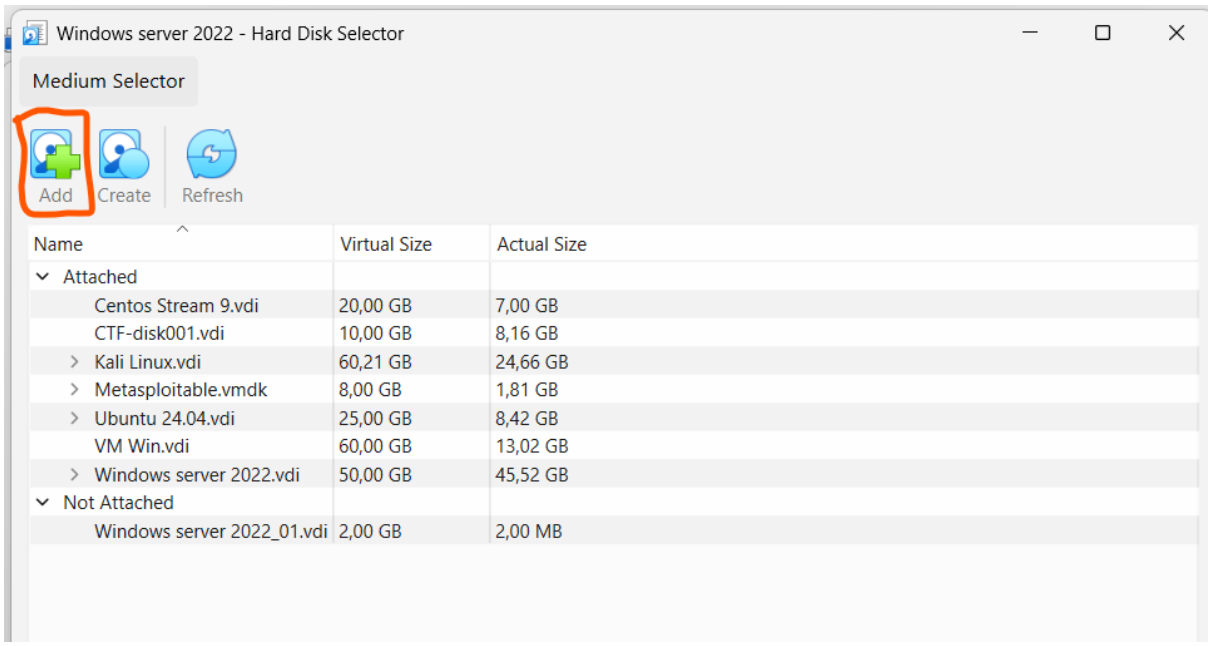


1.3 Click on Create Move slider To 2GB

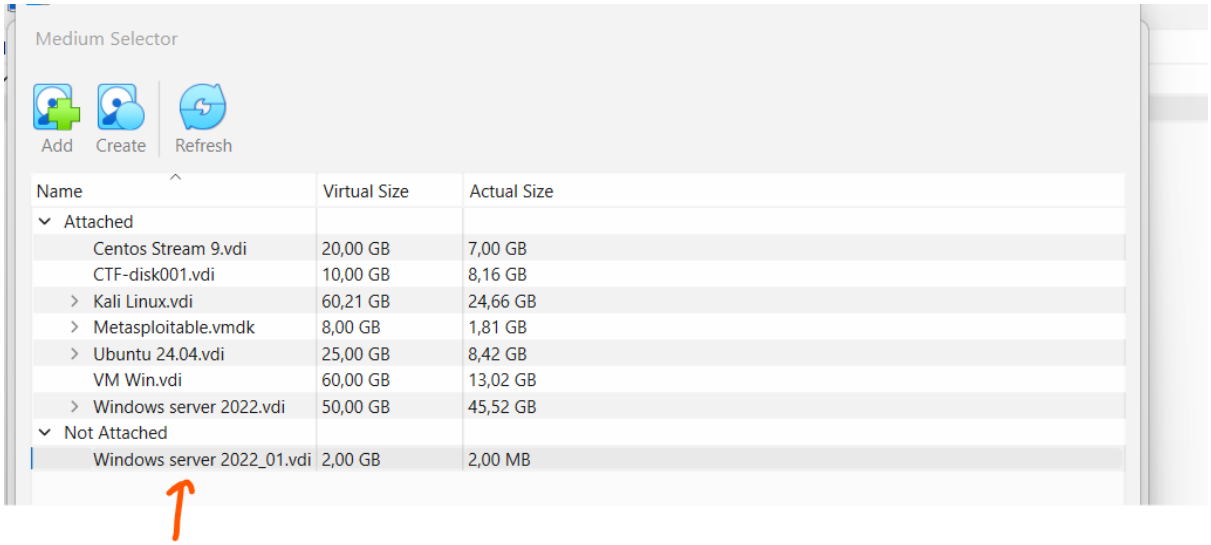
1.4 Change the disk name to 01.vdi then click on Finish



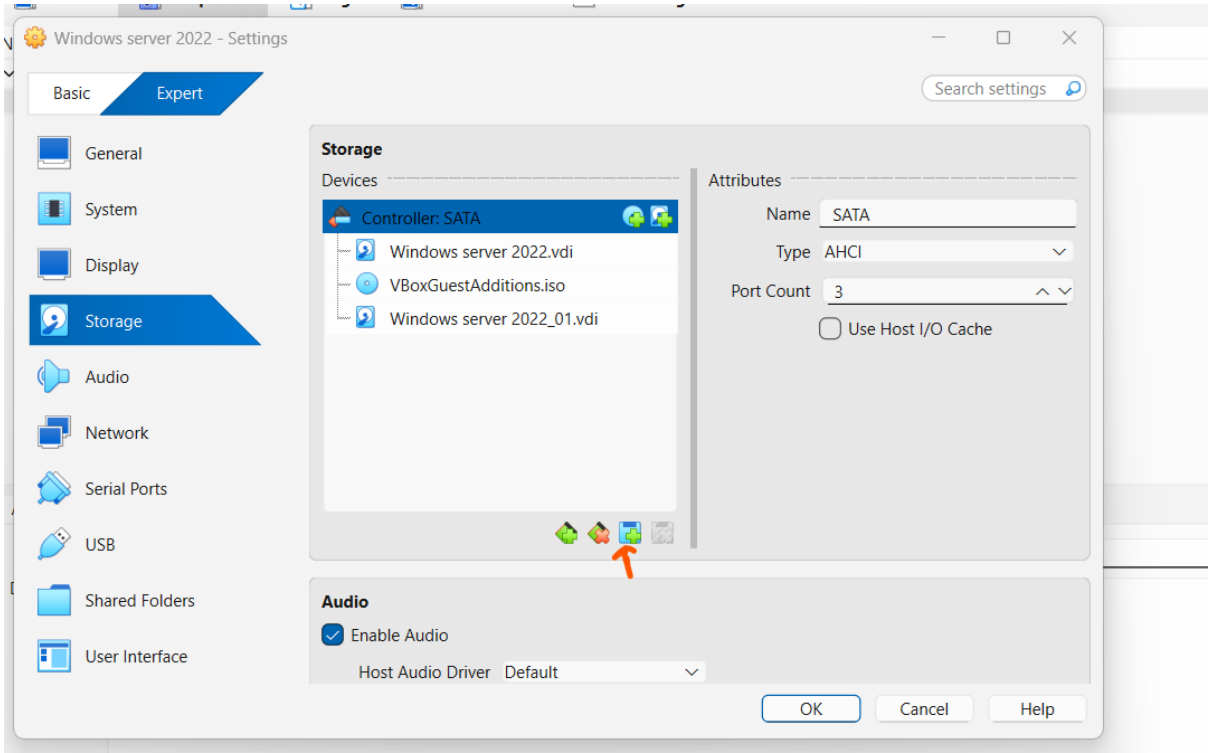
1.5 Click on Add



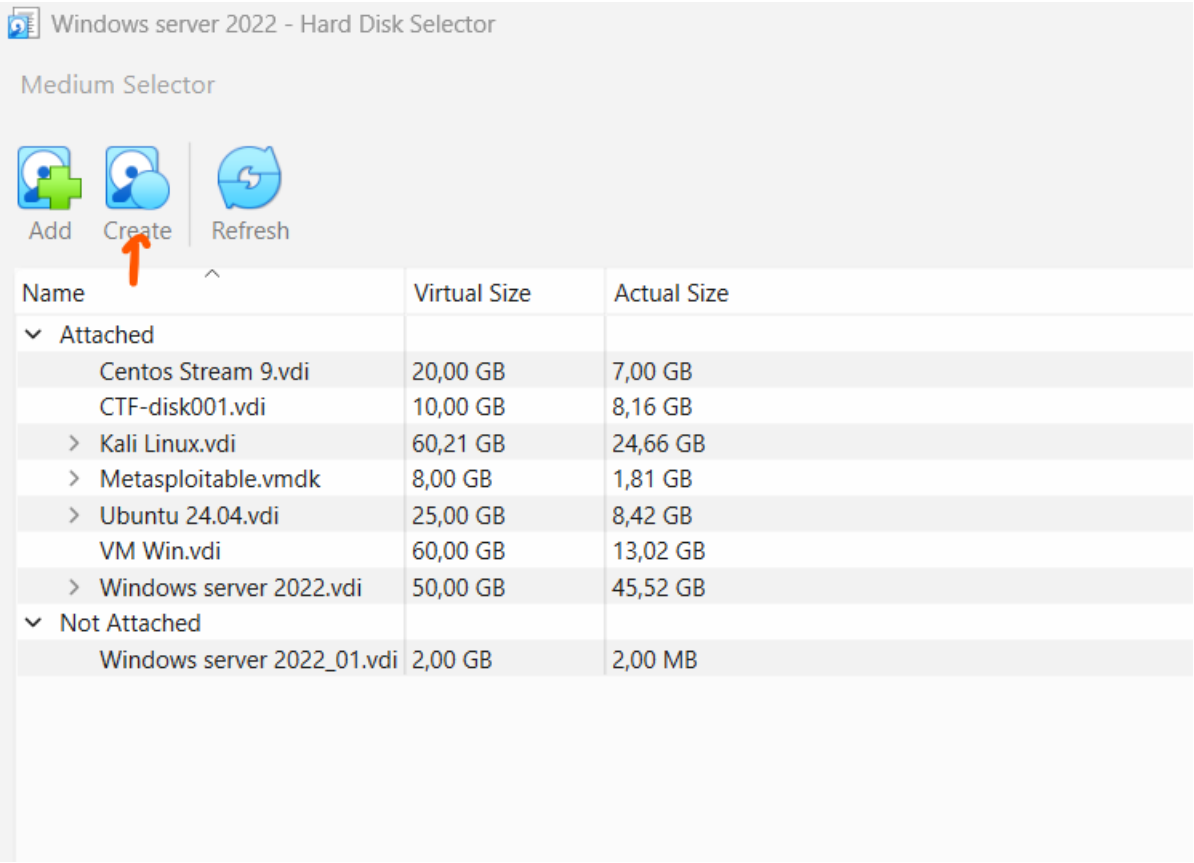
1.6 Select the 01.vhd and click on Open



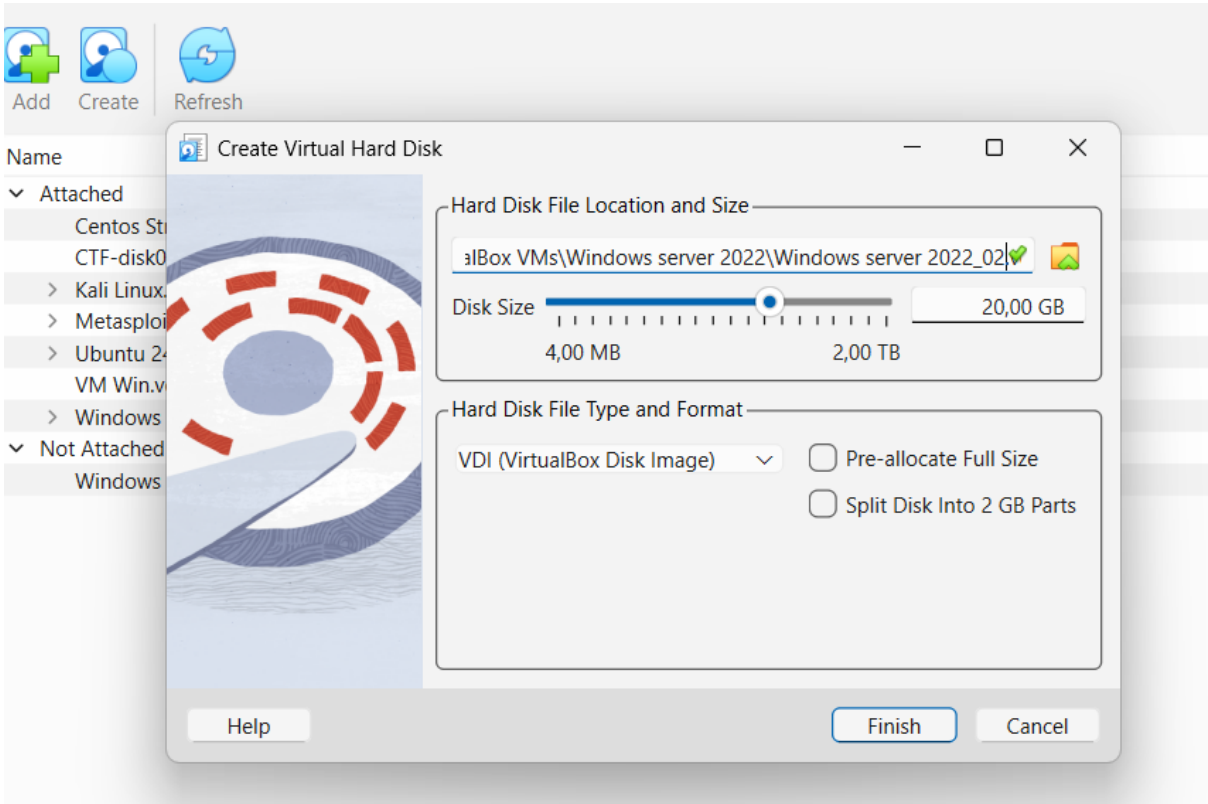
1.7 Now,again, to create a 20GB disk, click on AddAttachment



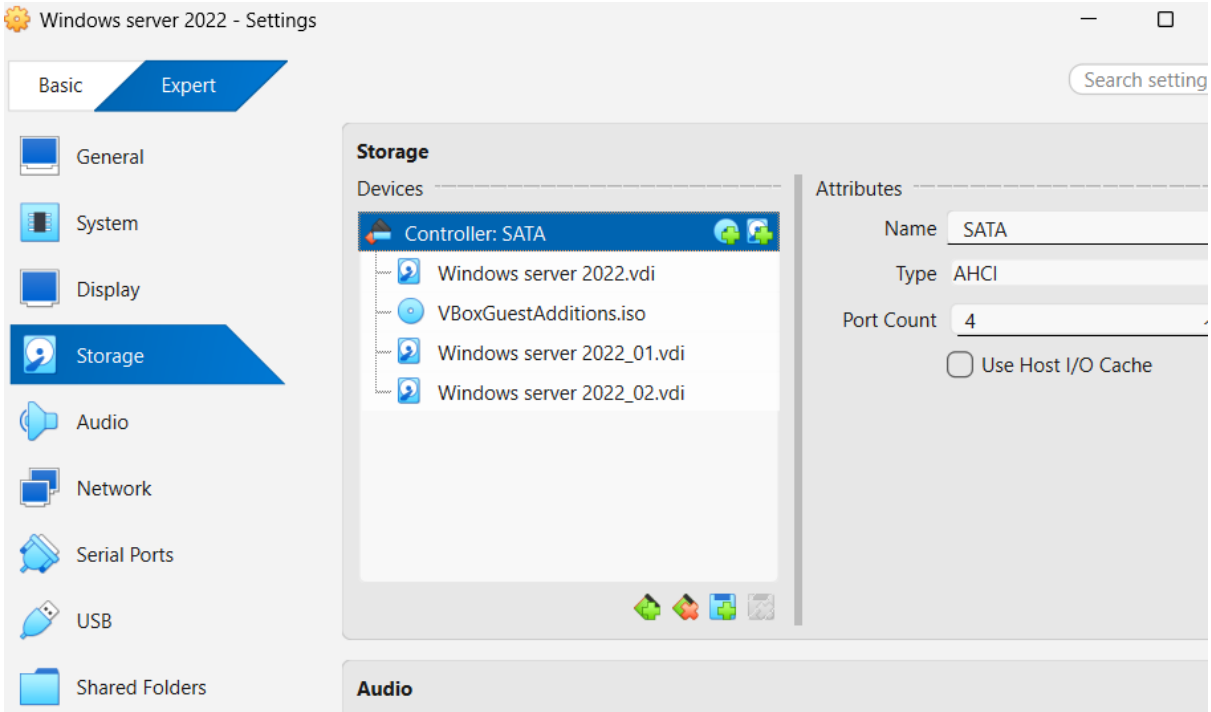
1.8 Select the Create button



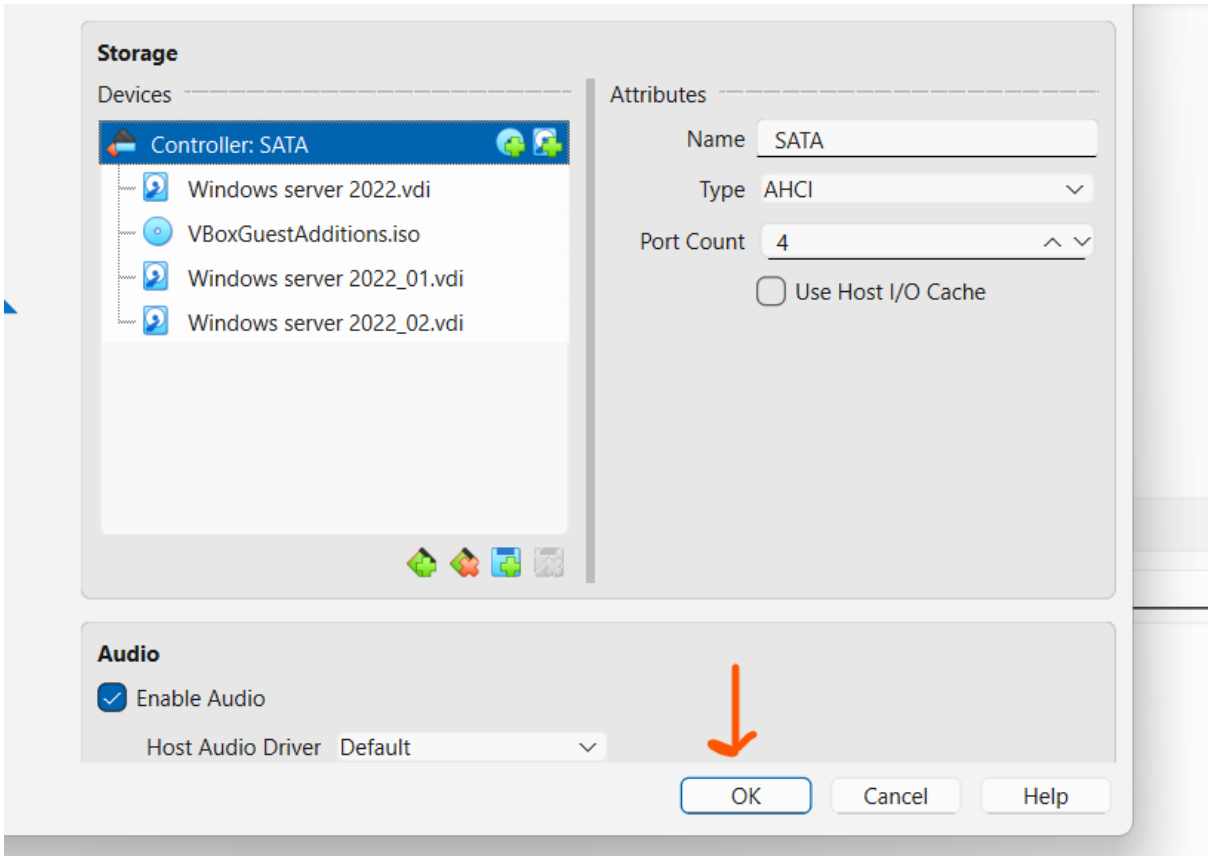
1.9 Change the disk name to 02.vdi, set the size to 20GB, and click Finish



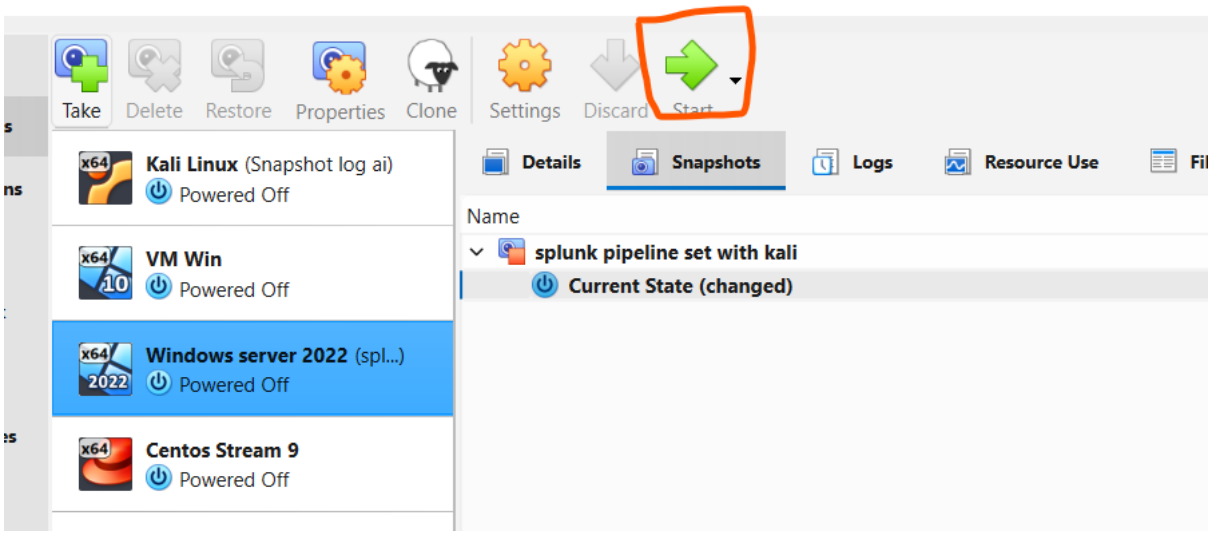
1.10 Select the 02.vhd and click on Choose



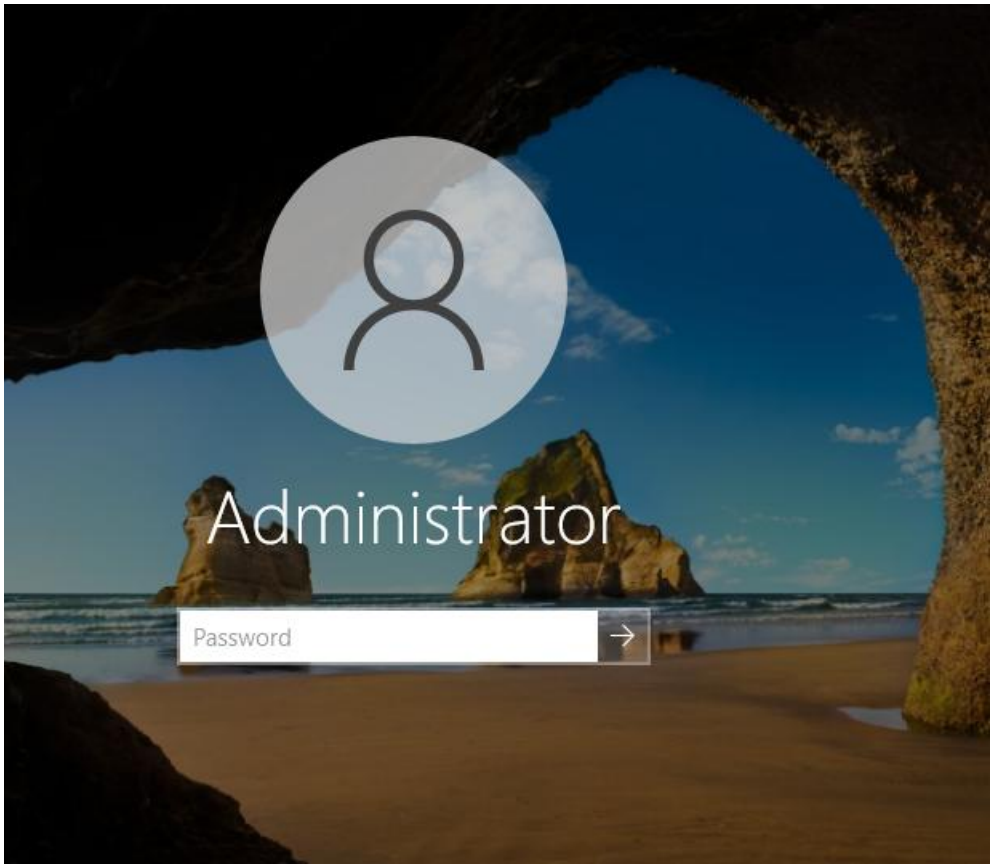
1.11 Click ok



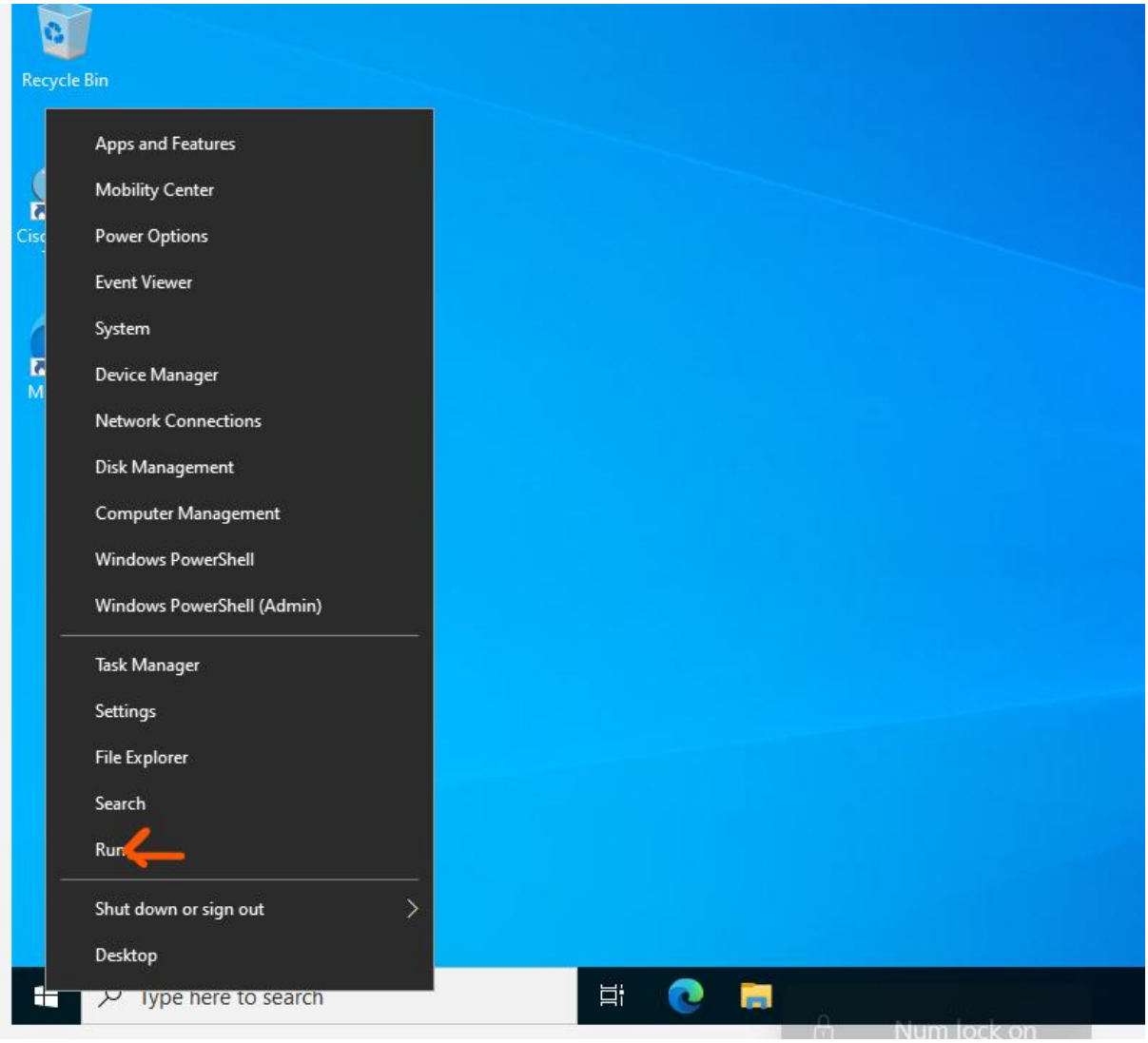
1.12 Start the Virtual machine



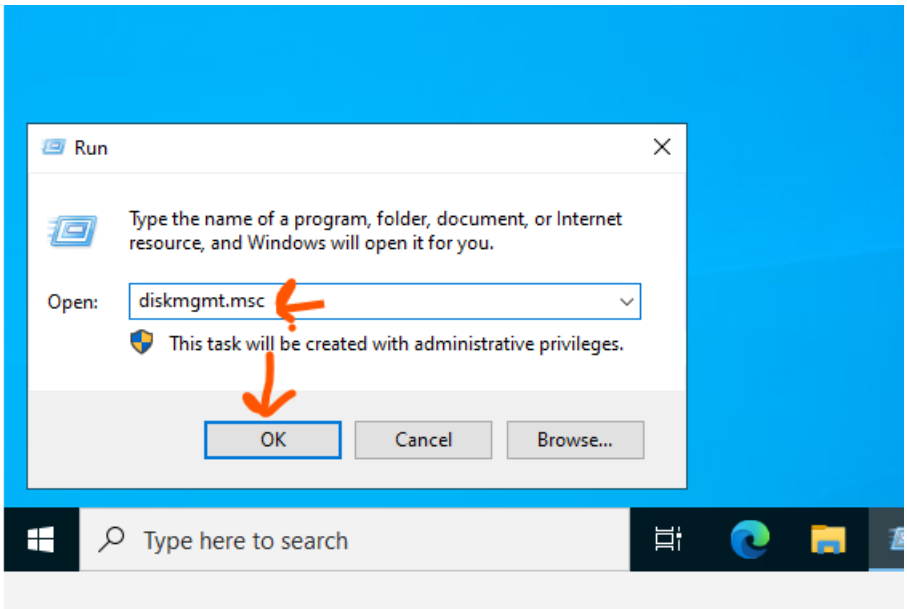
1.13 Log in as administrator



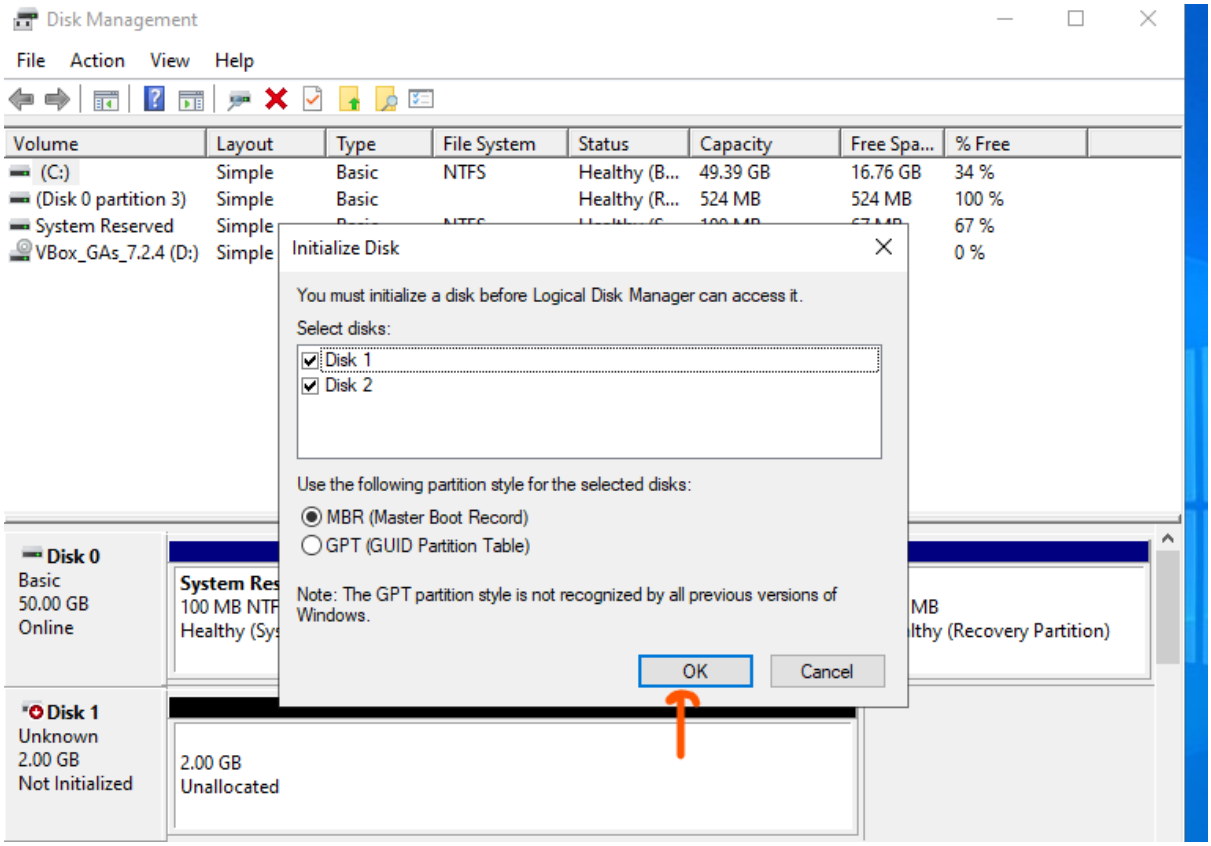
1.14 Right-click on Start and click on Run



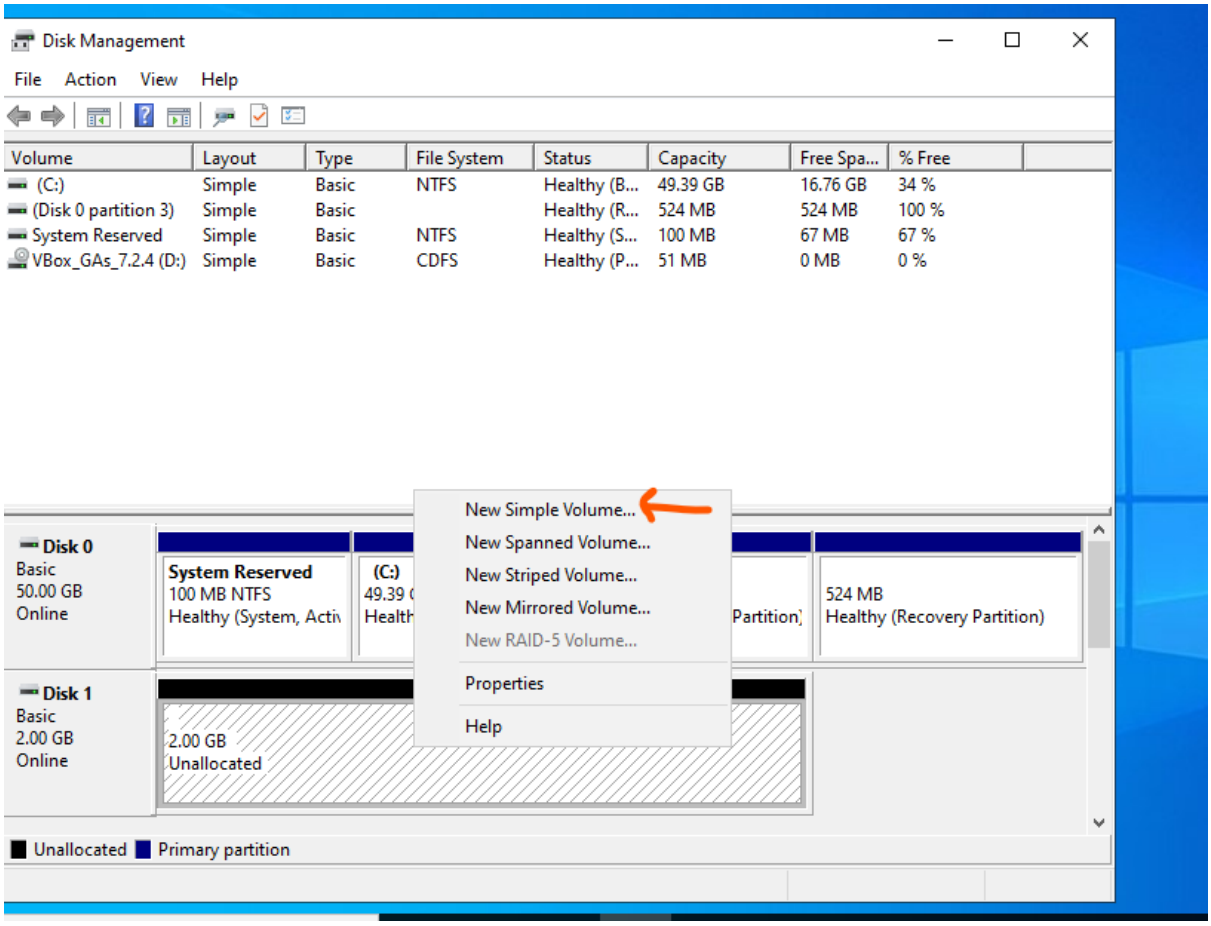
1.15 Type diskmgmt.msc and click on OK



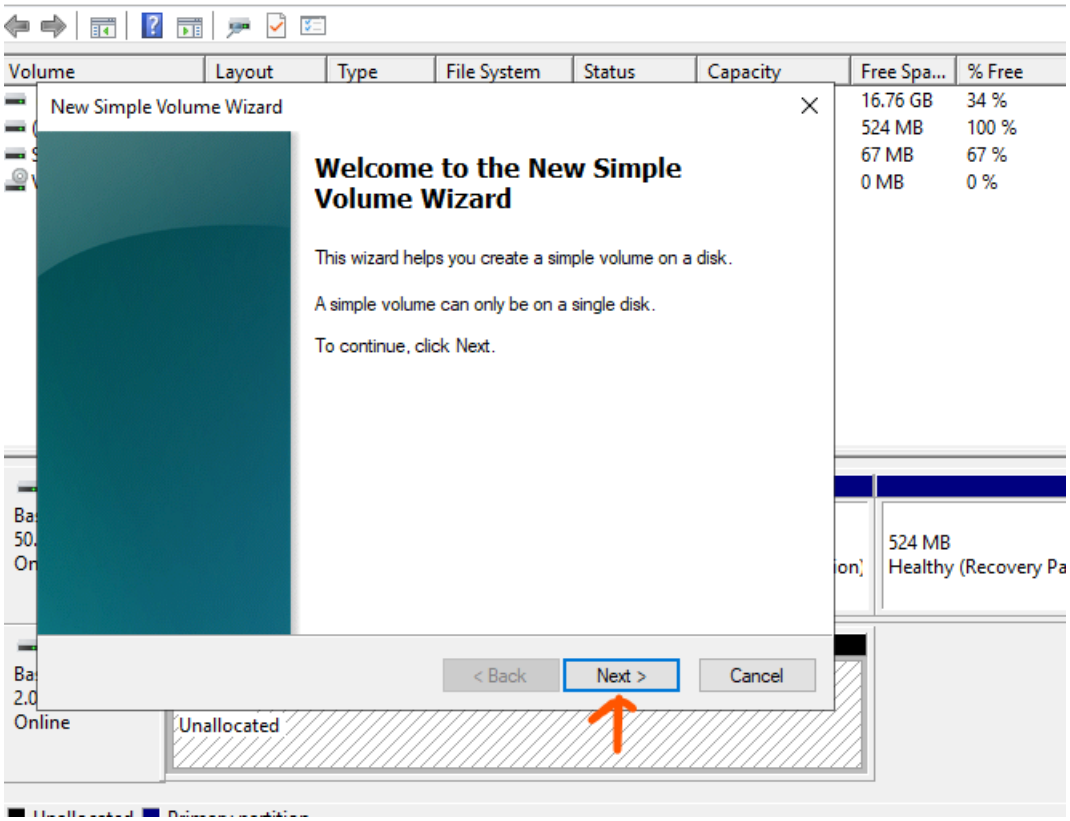
1.16 Click on OK



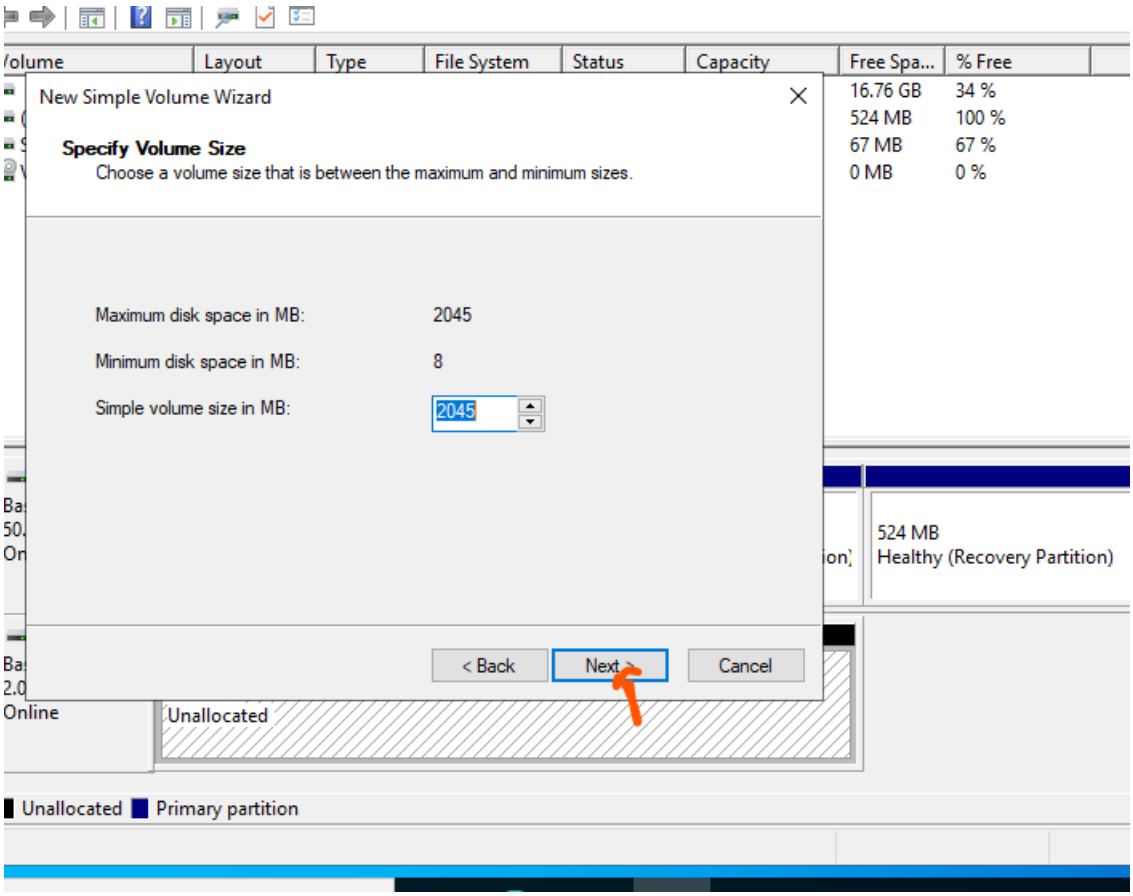
1.17 Right- click on Disk1 and select the NewSimpleVolume



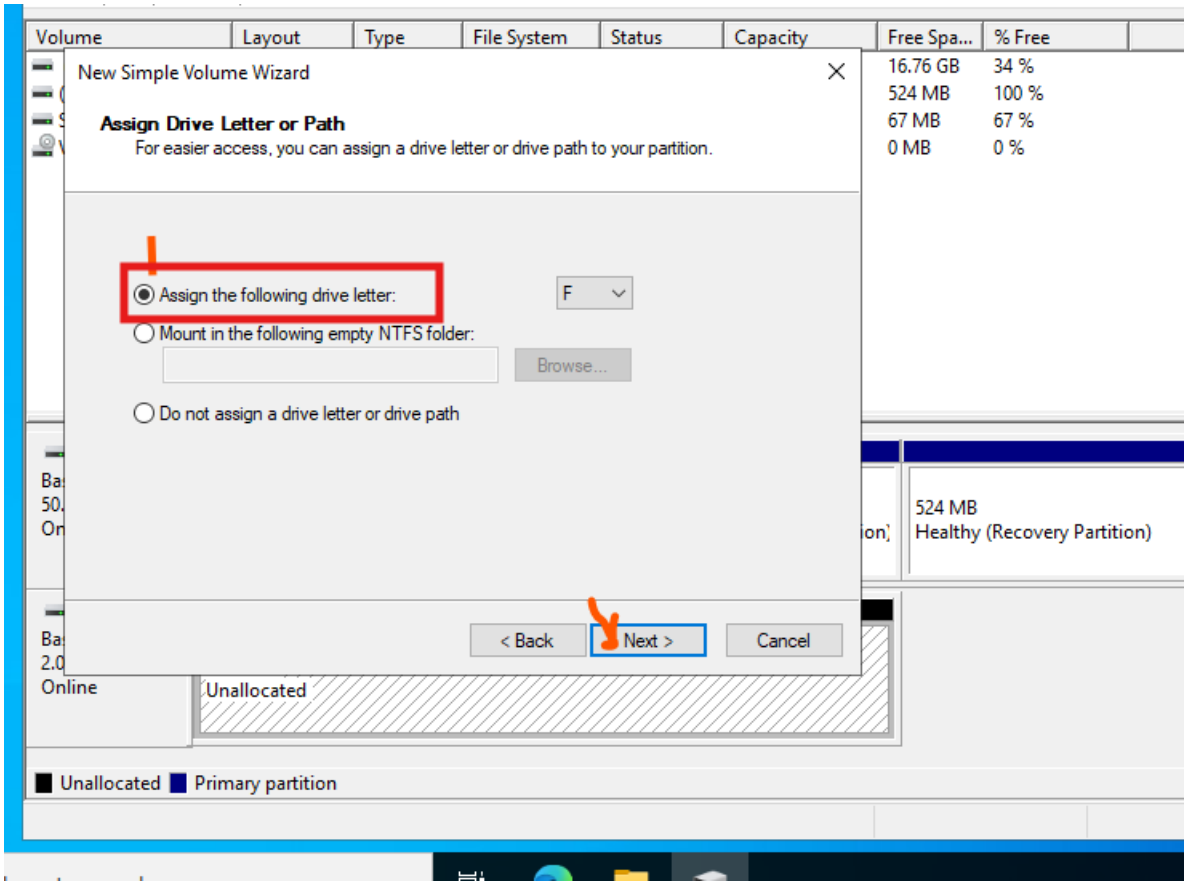
1.18 ClickonNext



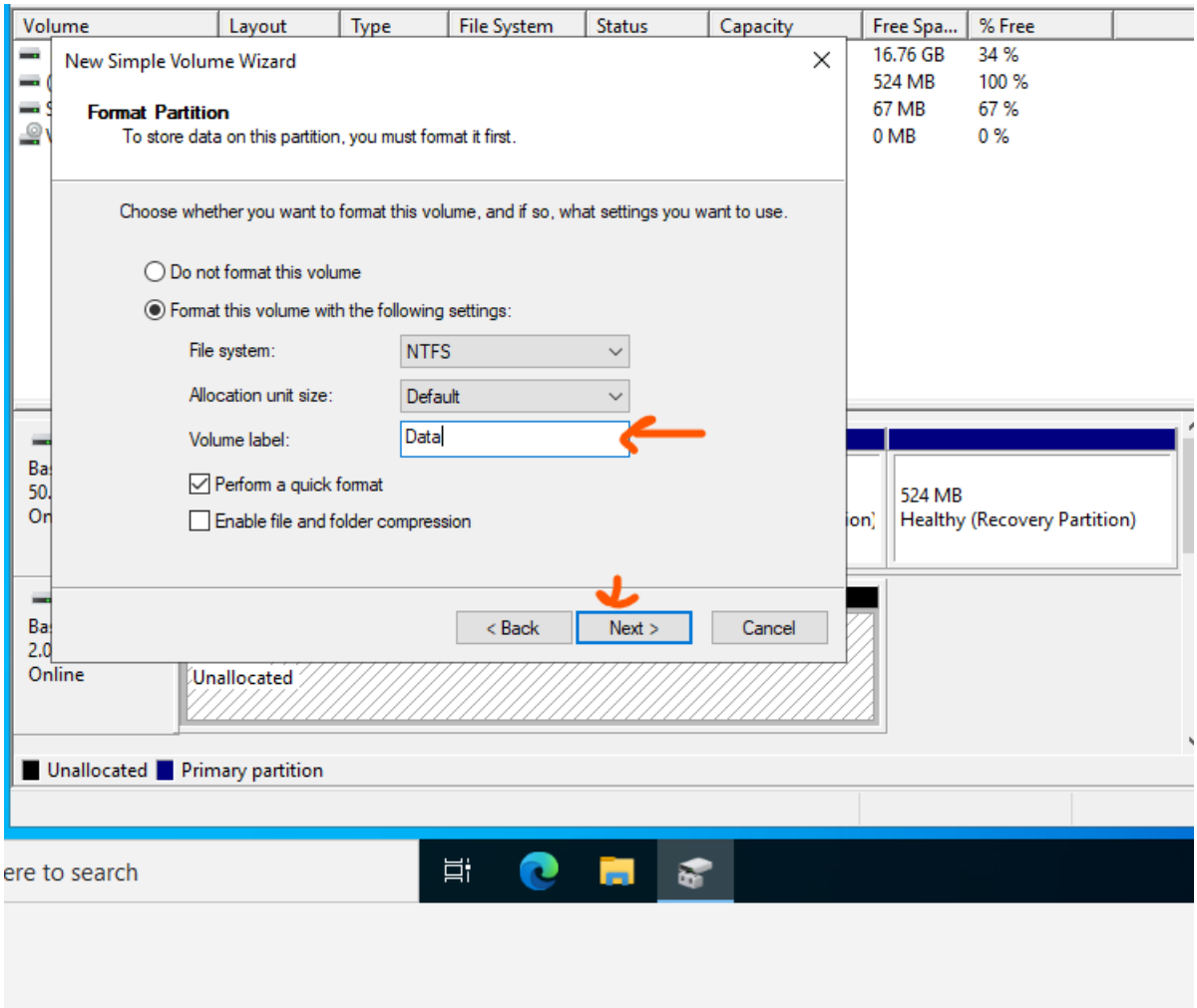
1.19 Click on Next



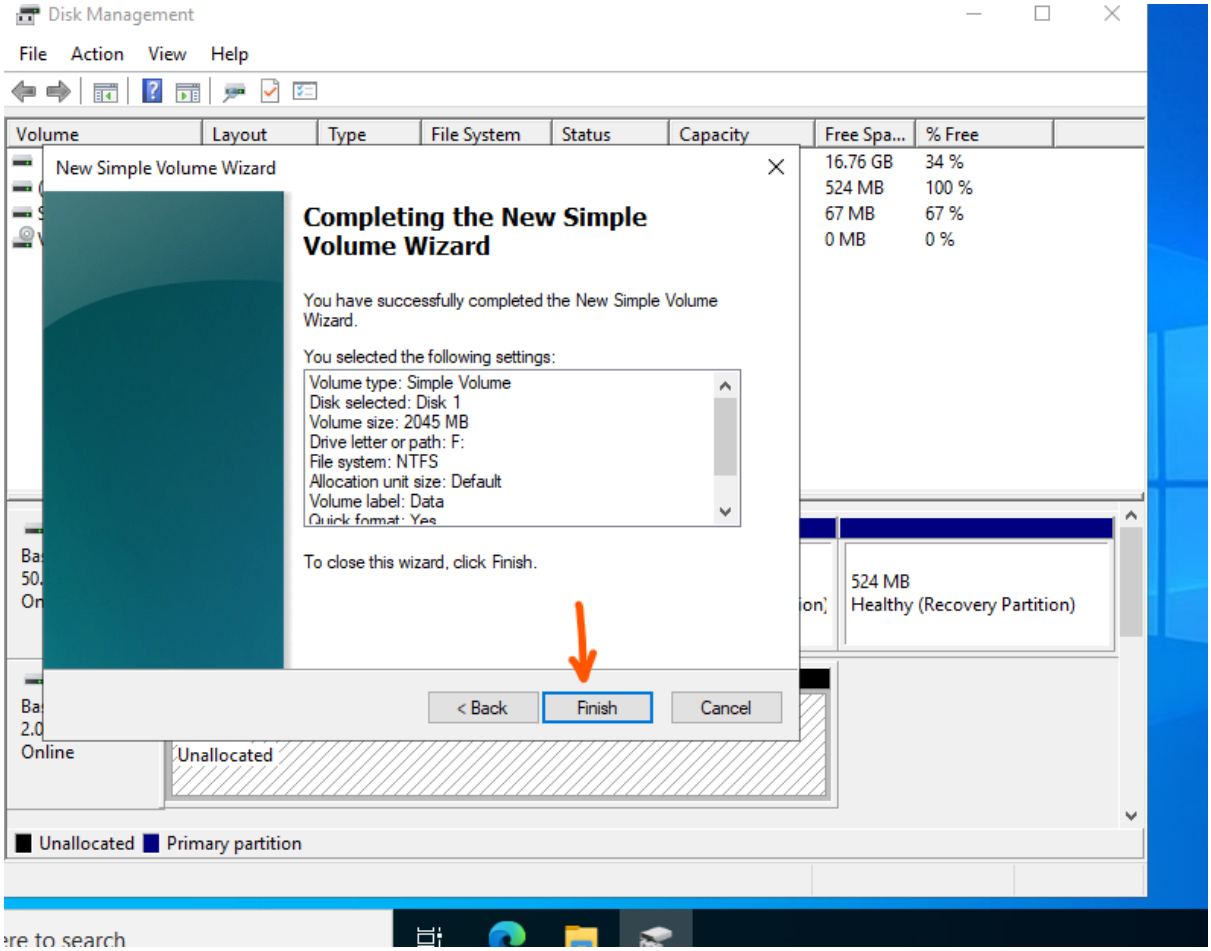
1.20 Check the radio button of Assign the following drive letter and click Next



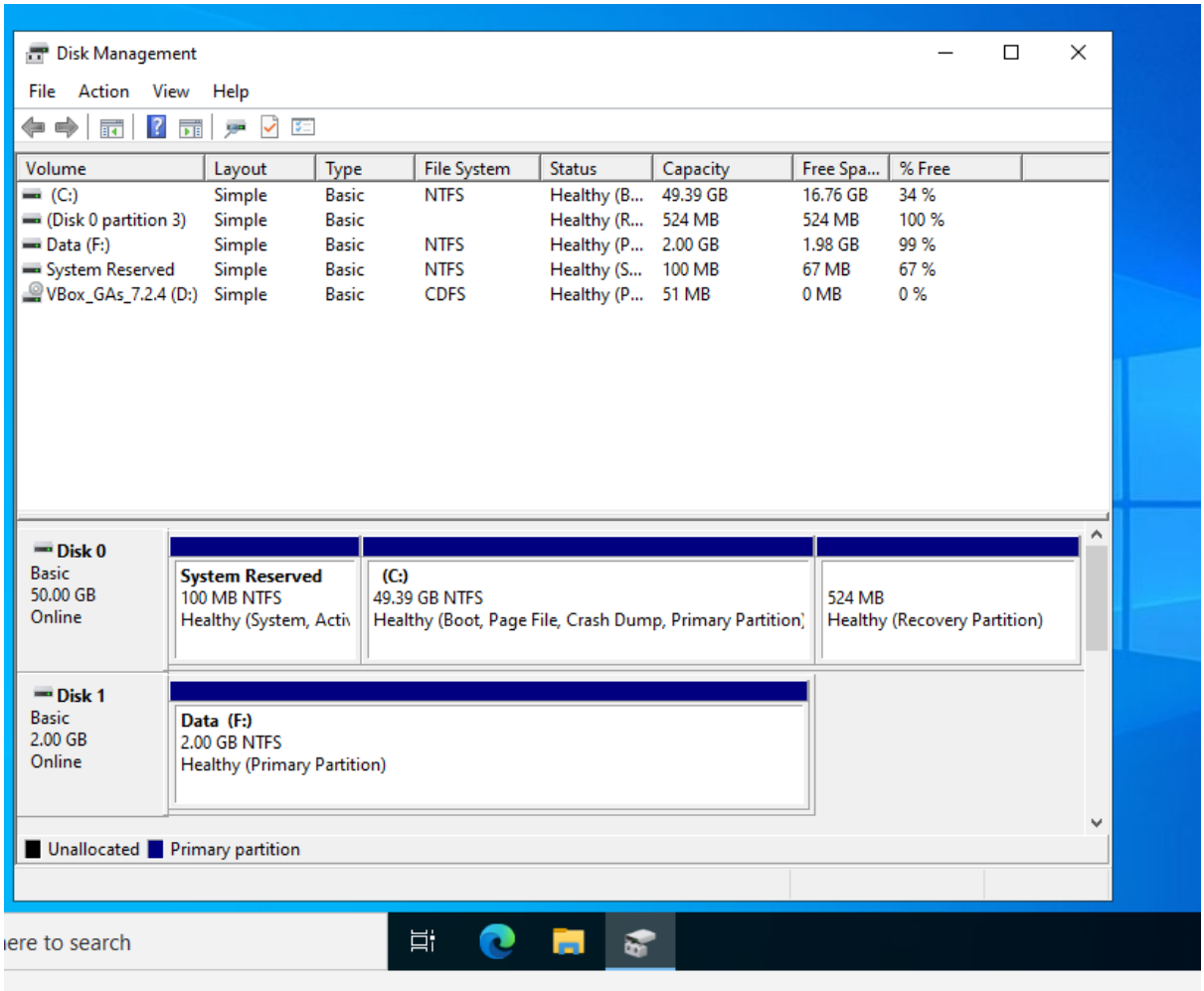
1.21 Write the Volume label as Data and click on Next



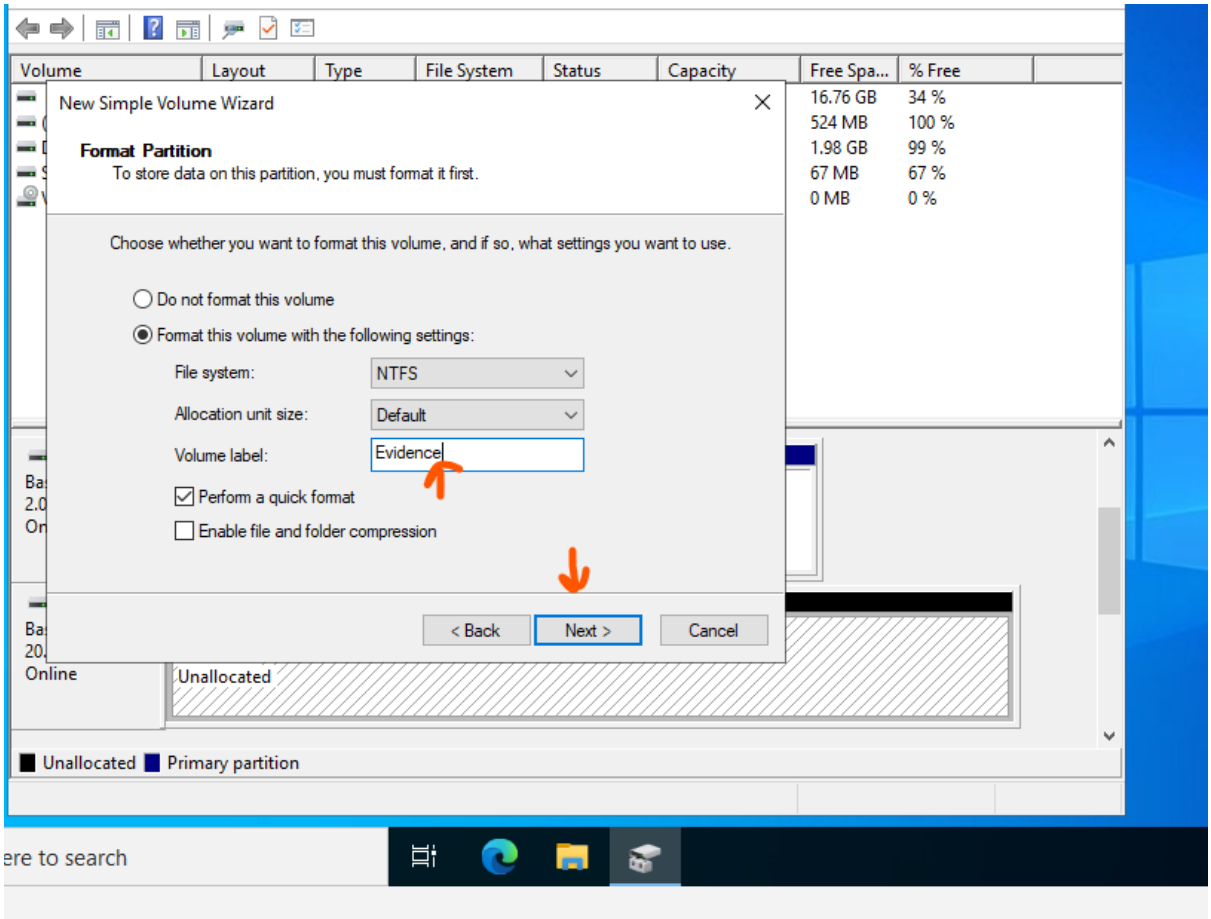
1.22 Click on Finish



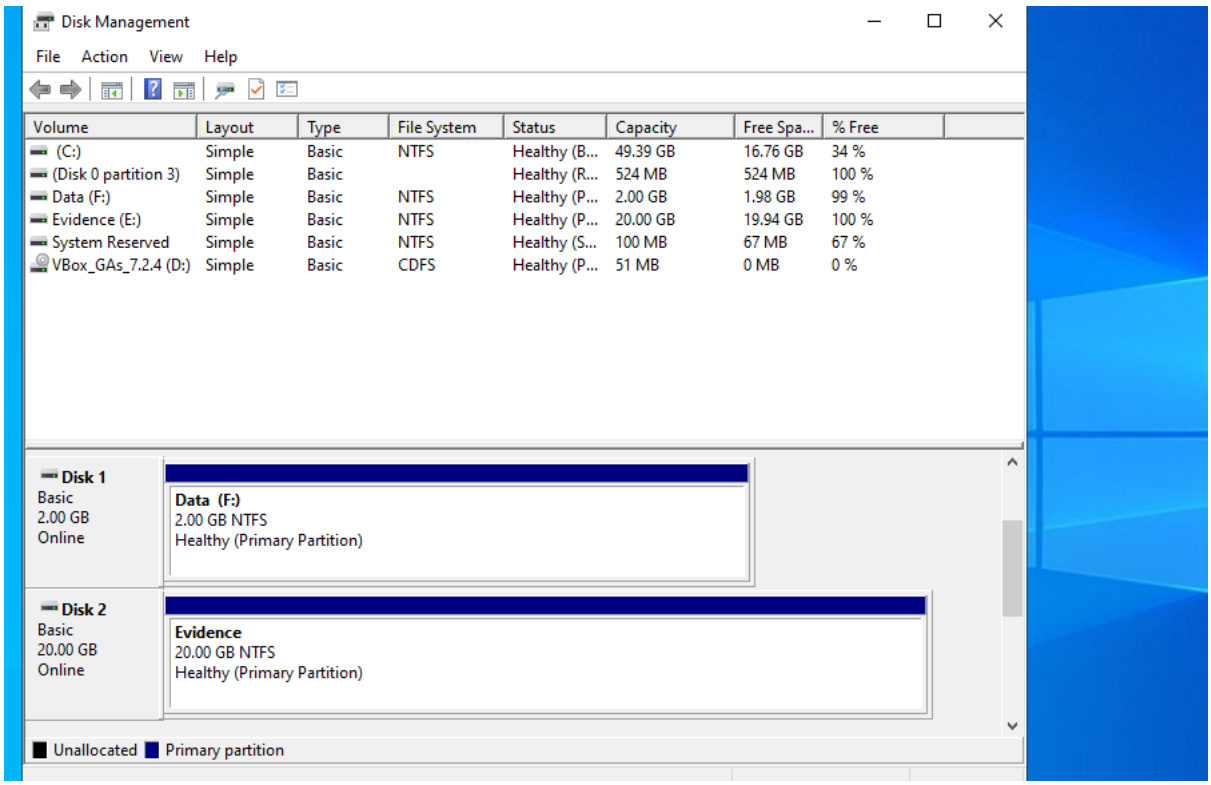
1.23 You can see that Disk1 is now a healthy partition.



1.24 Similarly , create a newpartition of 20 GB and label it as Evidence

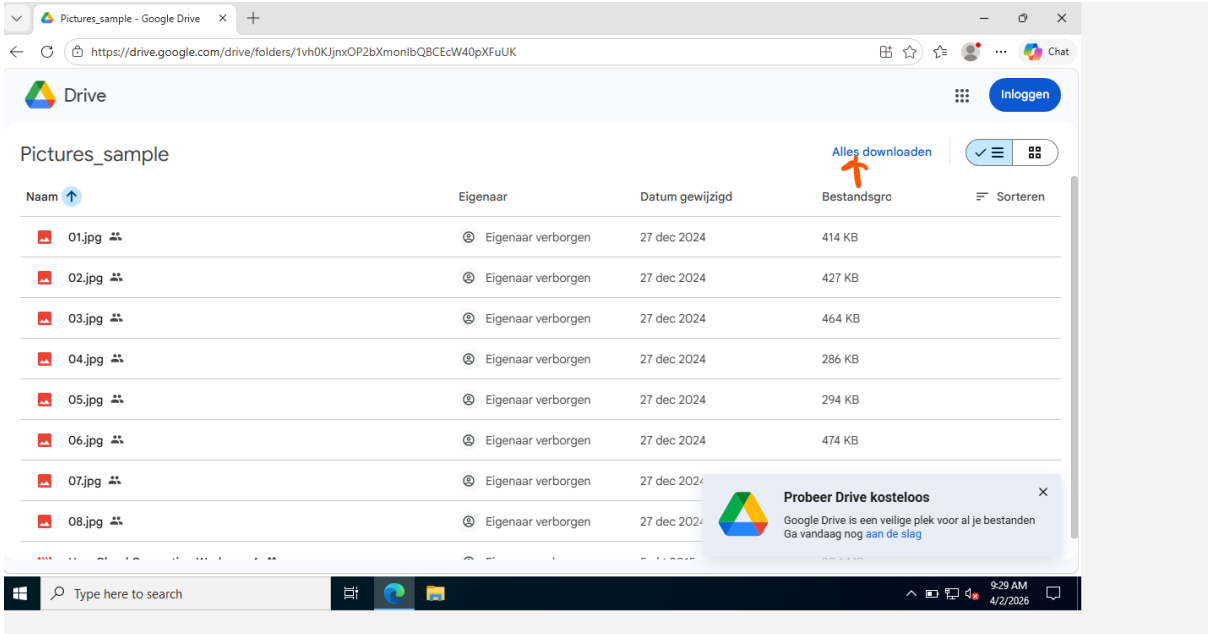


1.25 Now both disk1 and disk 2 have healthy partition

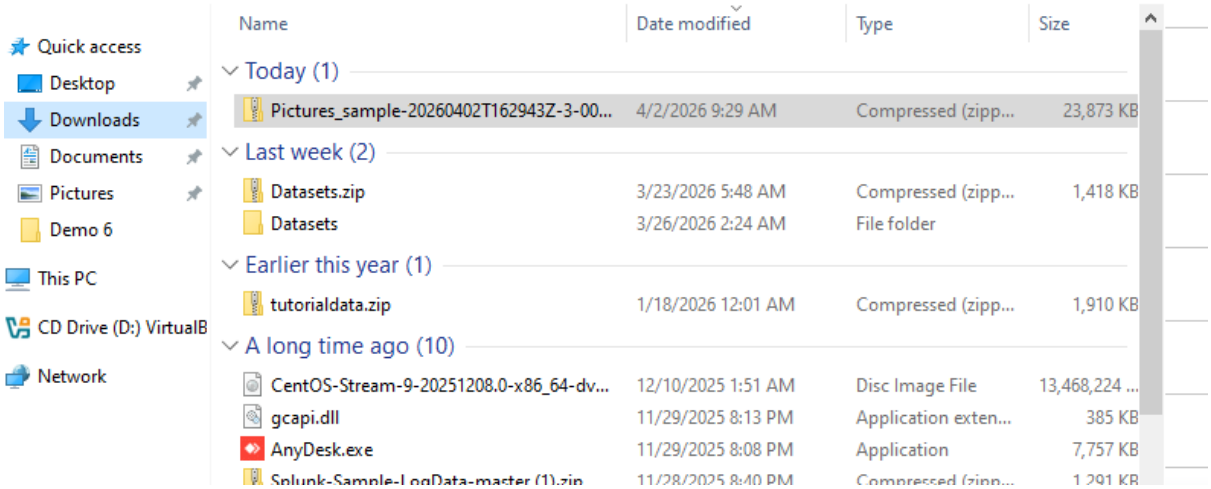


Step 2: Download and prepare the data to be tested

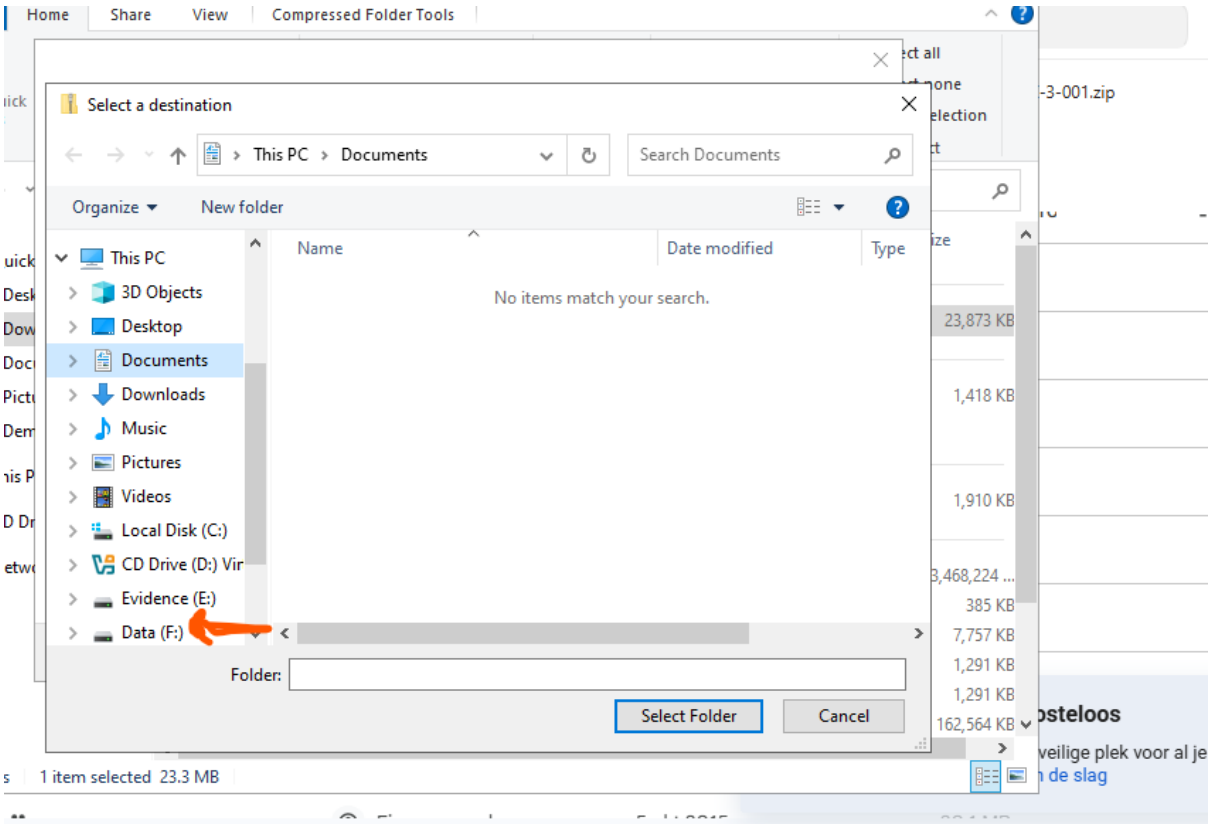
2.1 Download all files from the link given below:
https://drive.google.com/drive/folders/1vh0KJjnxOP2bXmonlbQBCEcW40pXFuUK?us p=sharin



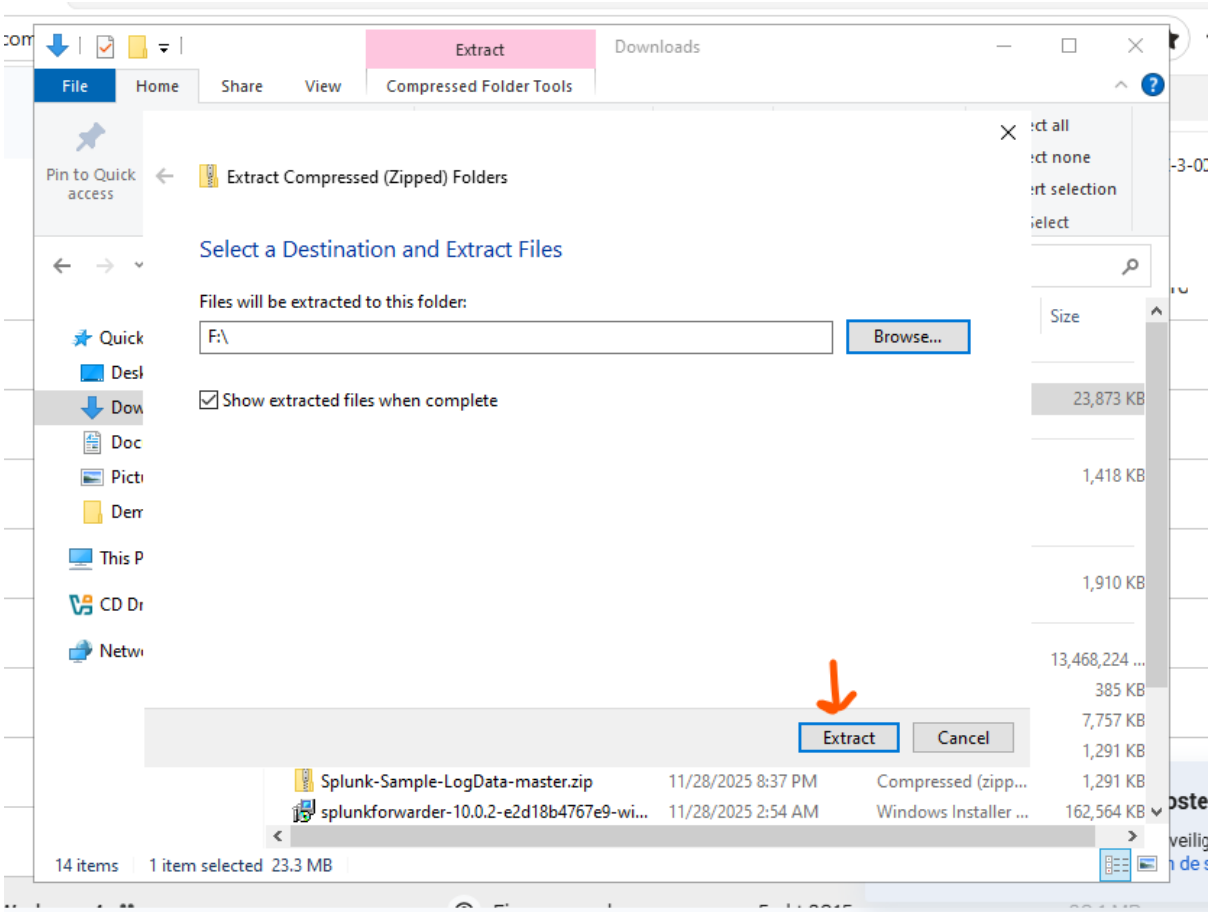
2.2 Right-click on the file and click on Extract all

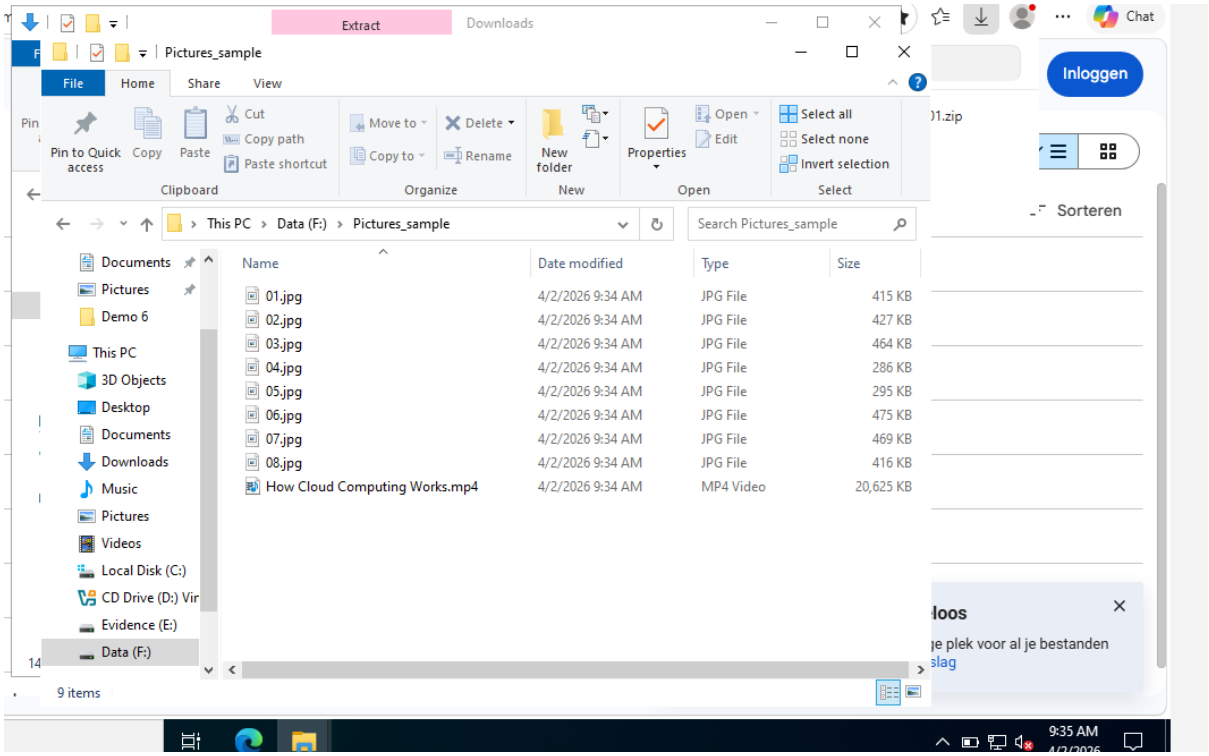


2.3 Select the Data drive and click on Select Folder



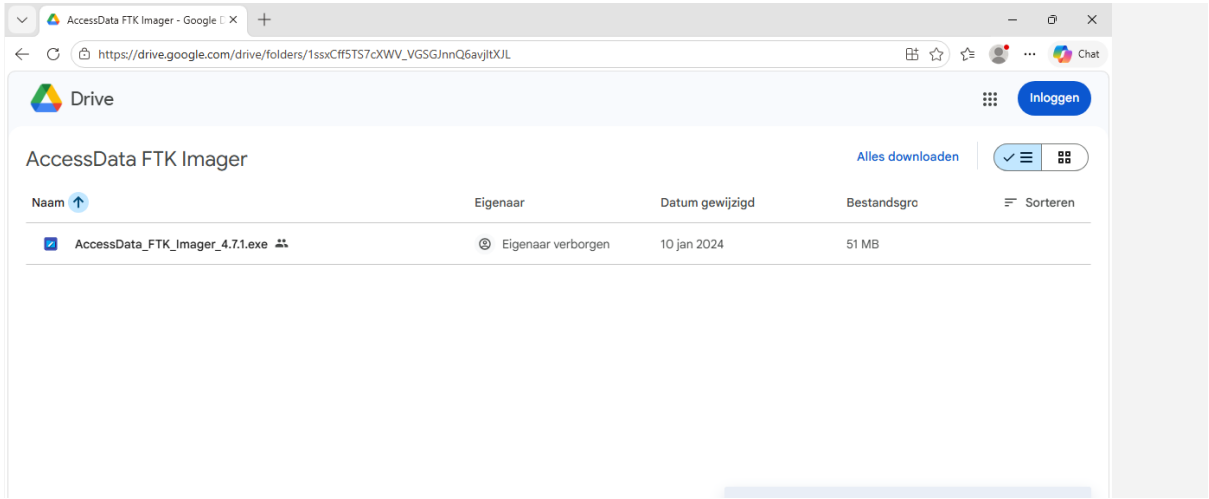
2.4 Click on Extract



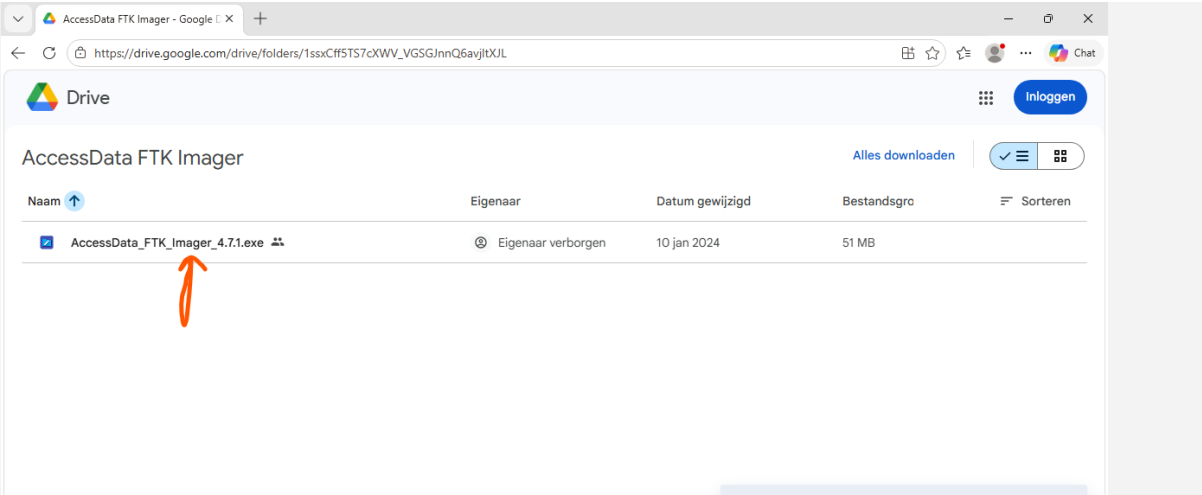


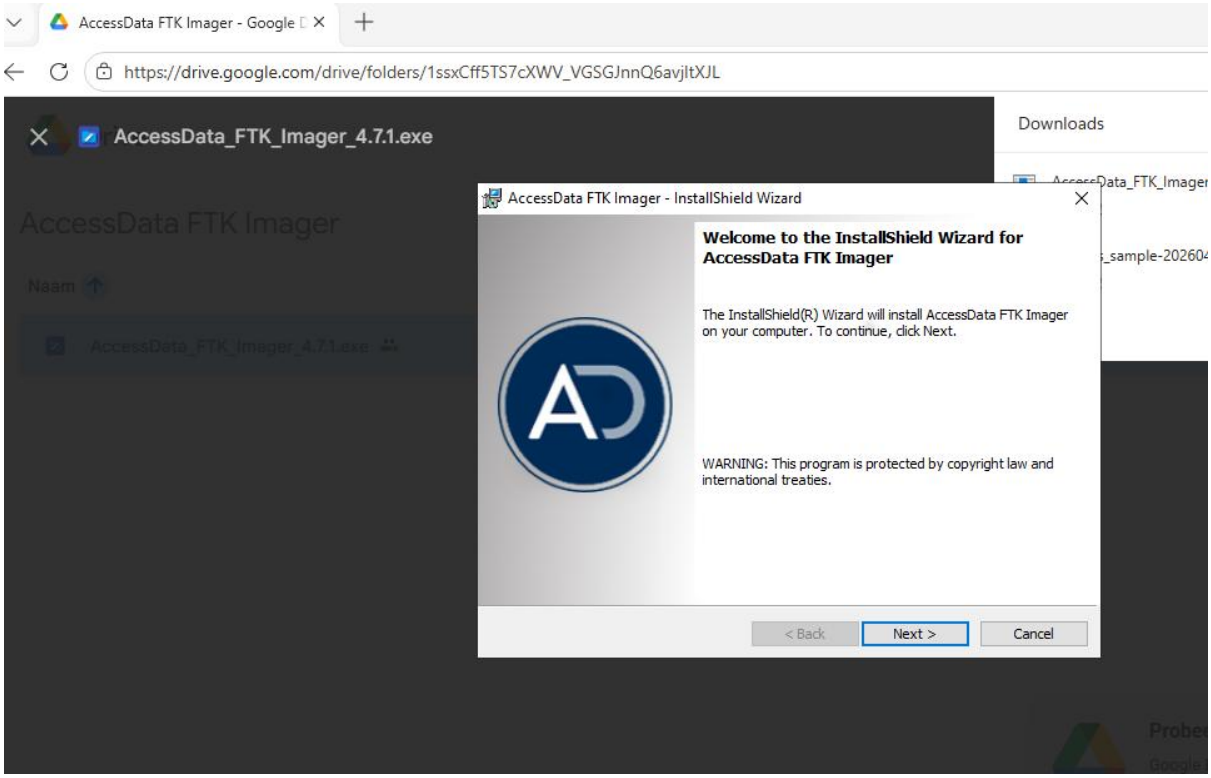
Step3:DownloadAccessdataFTKImager

3.1 Download Access data FTK Images from the following link: https://drive.google.com/drive/folders/1ssxCff5TS7cXWV_VGSGJnnQ6avjltXJL



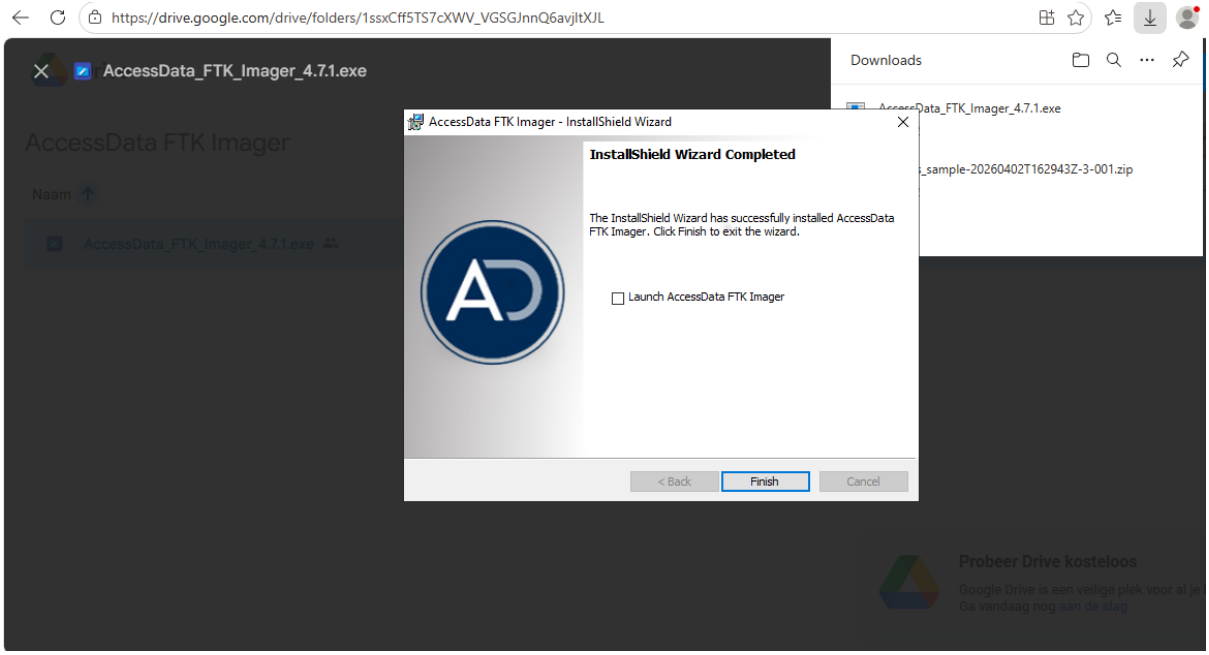
3.2 Double-click on the setup of FTK Imager and click Next





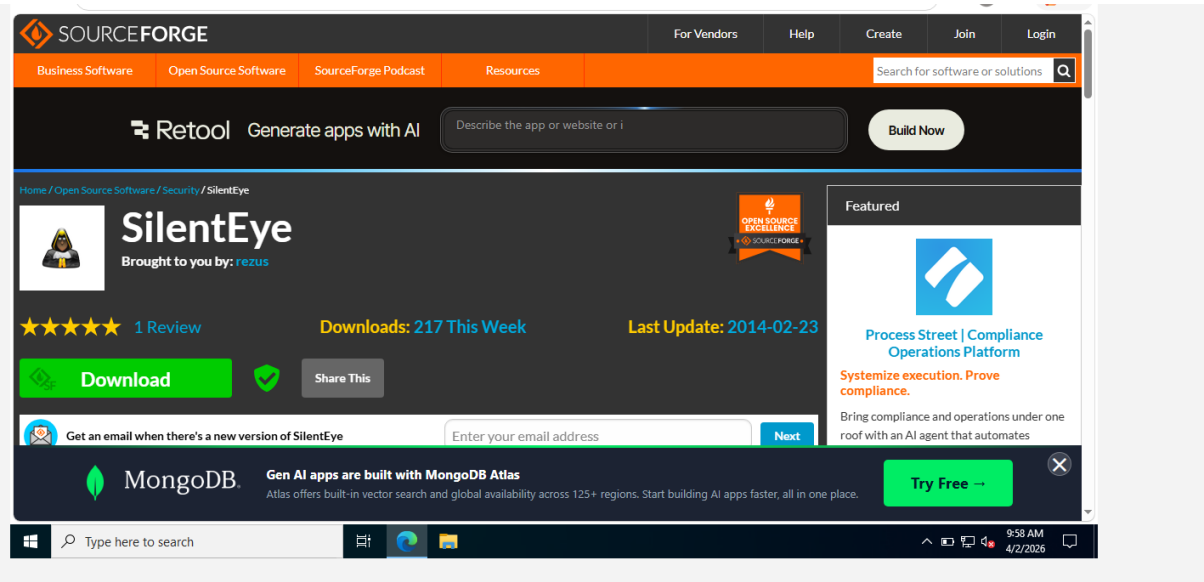
3.3 Accept the license agreement and click Next

3.4 Uncheck the radio button and click on Finish

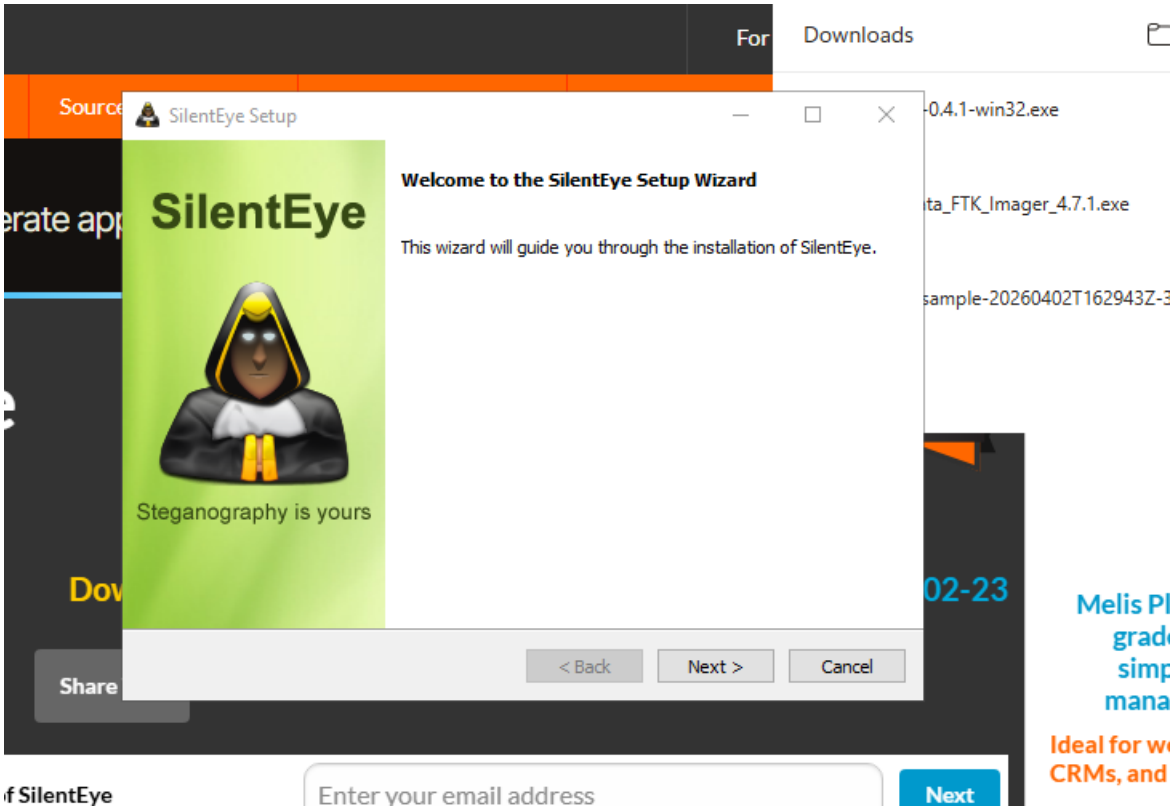


Step 4: Hide text in the image using Steganography

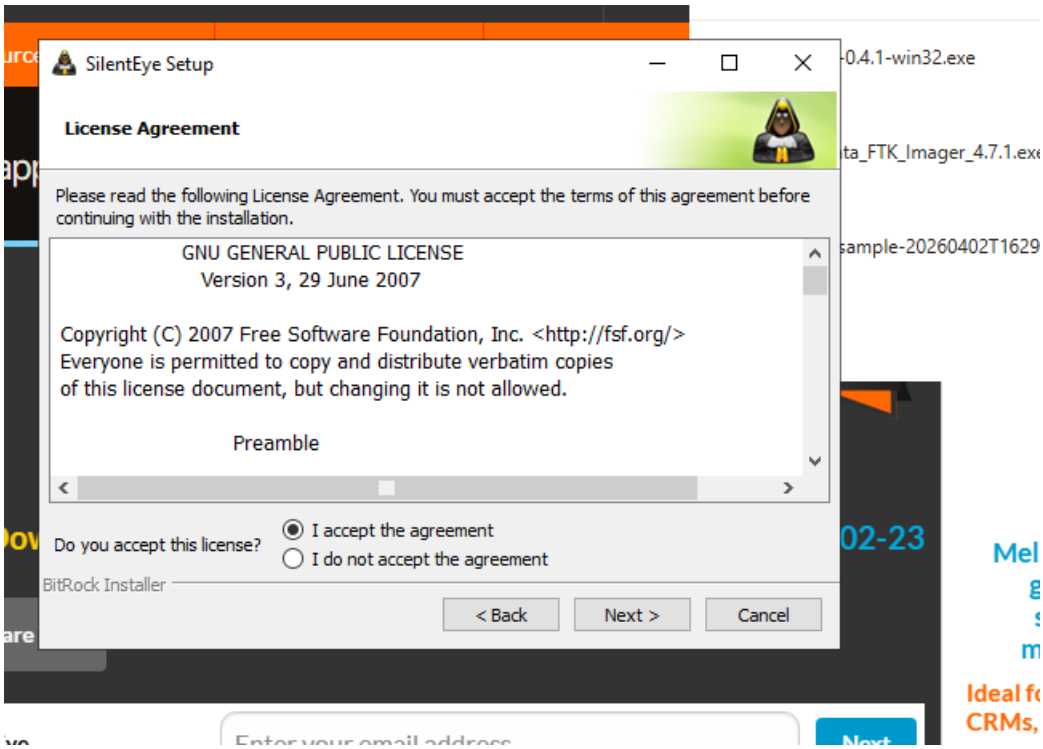
4.1DownloadSilentEyefromthebelowlink:
https://sourceforge.net/projects/silenteye/



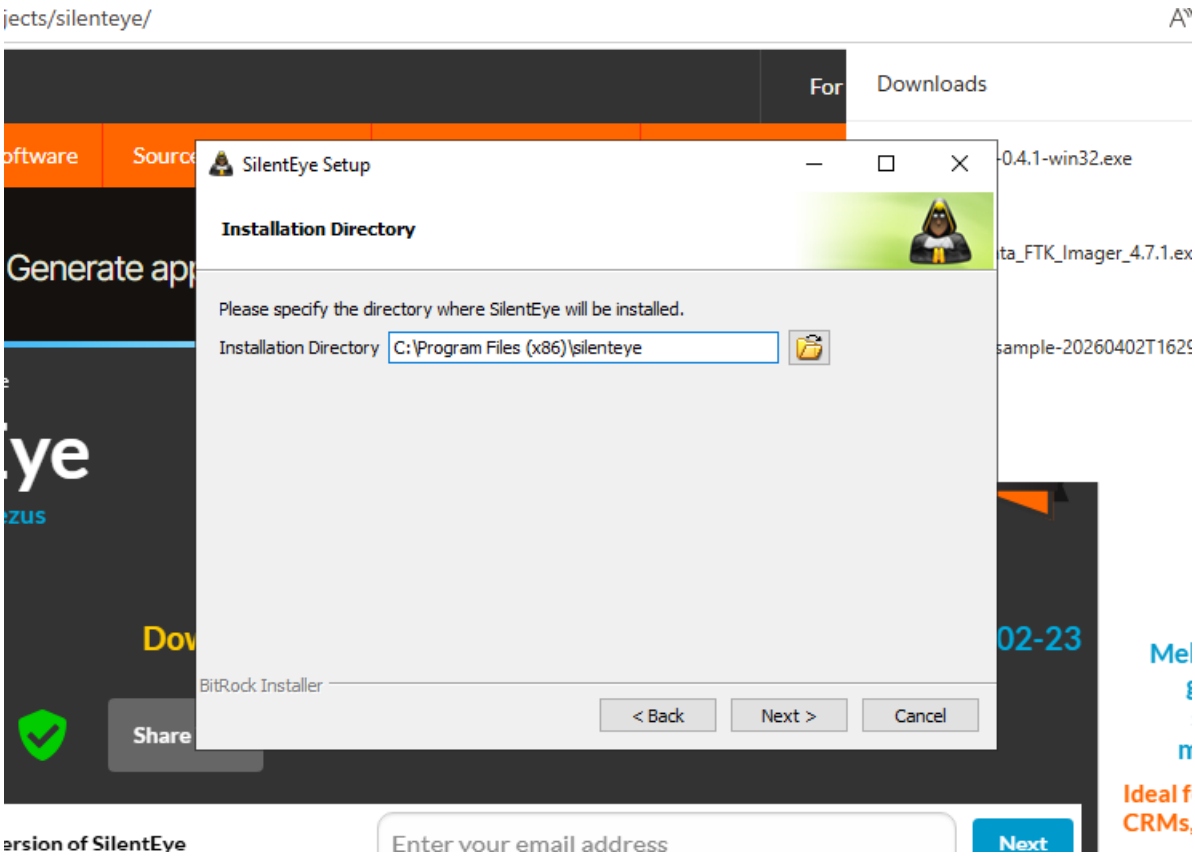
4.2 Click on Next



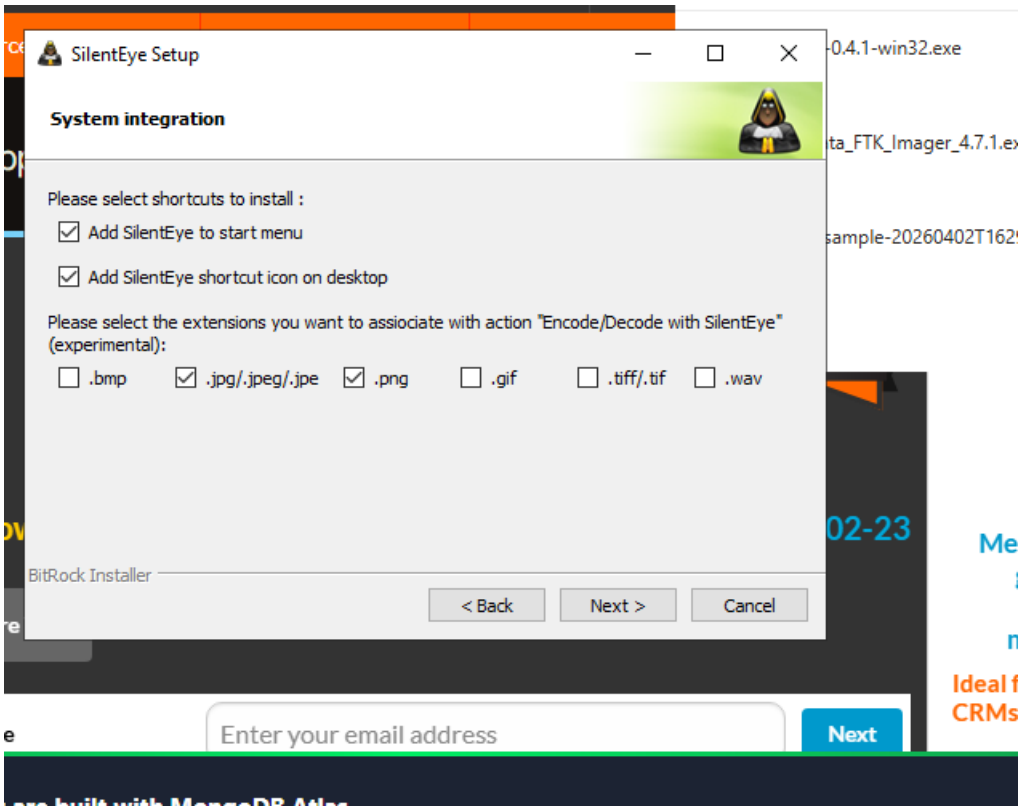
4.3 Accept the Agreement and click on Next



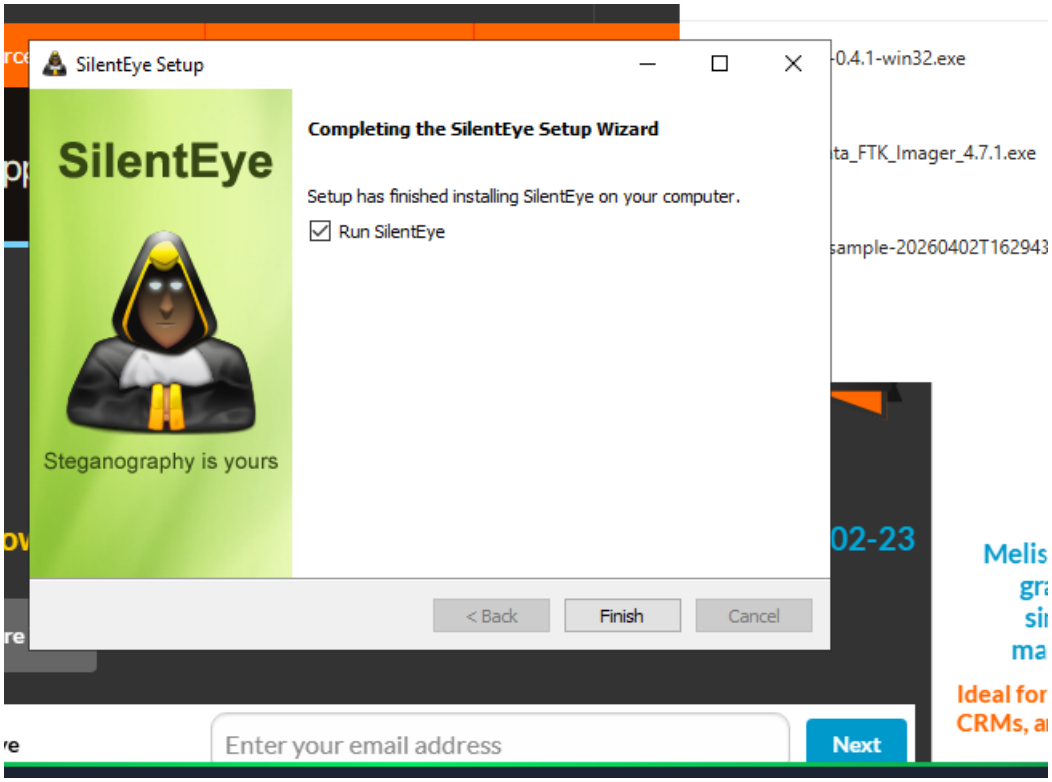
4.4 ClickonNext



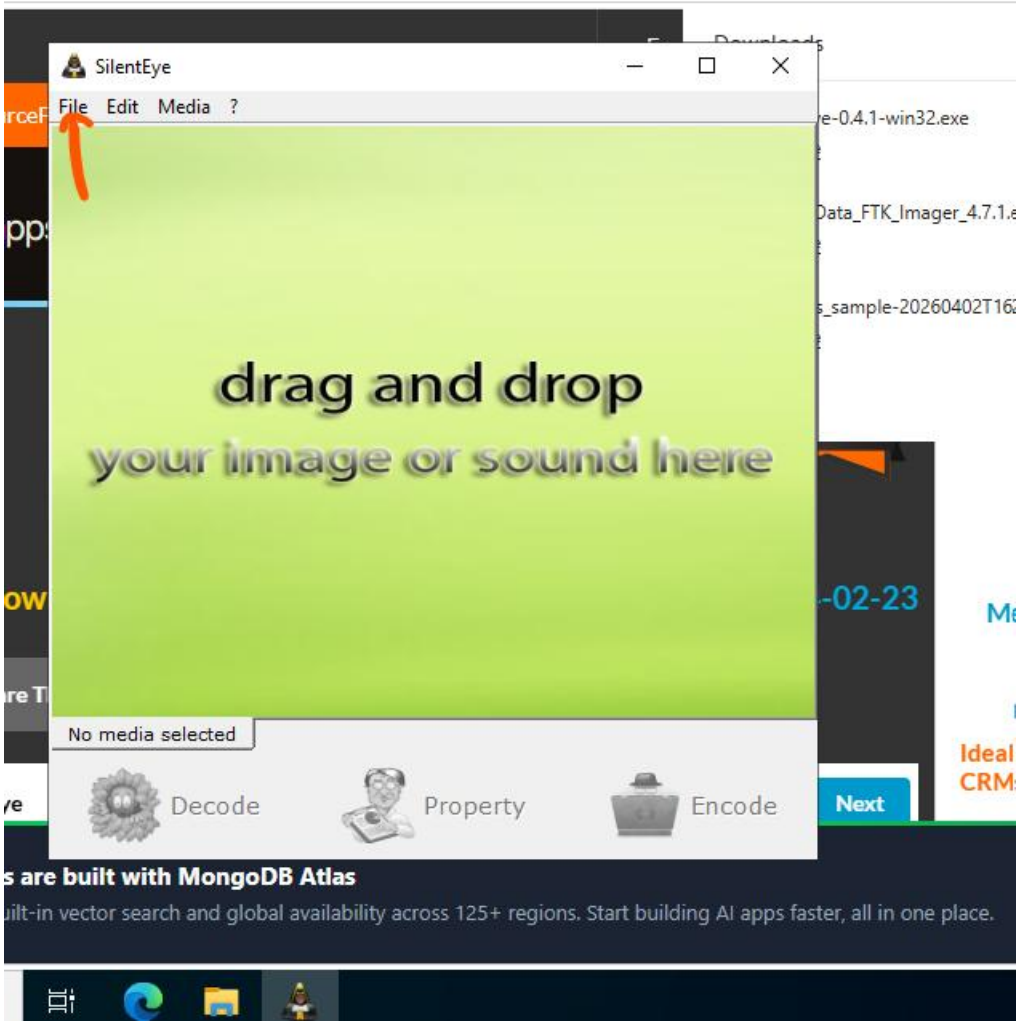
4.5 Click on Next



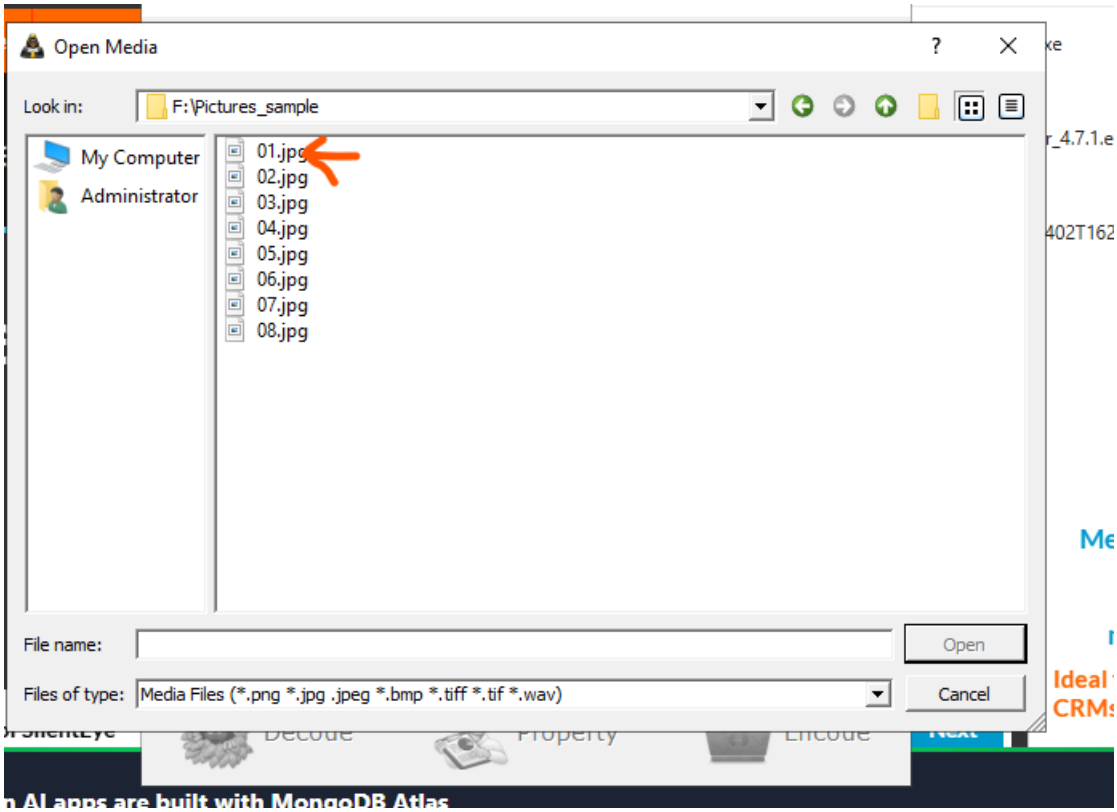
4.6 Click Finish



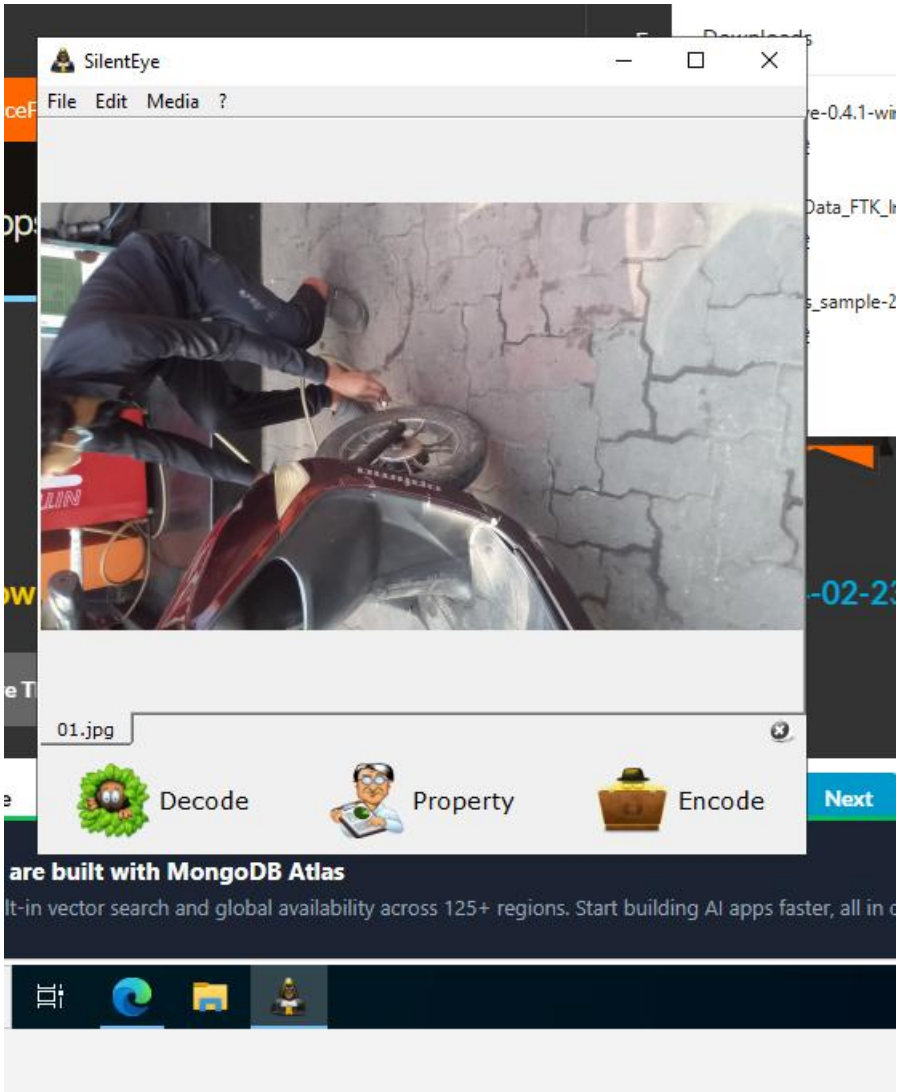
4.7 Click on File and select Open



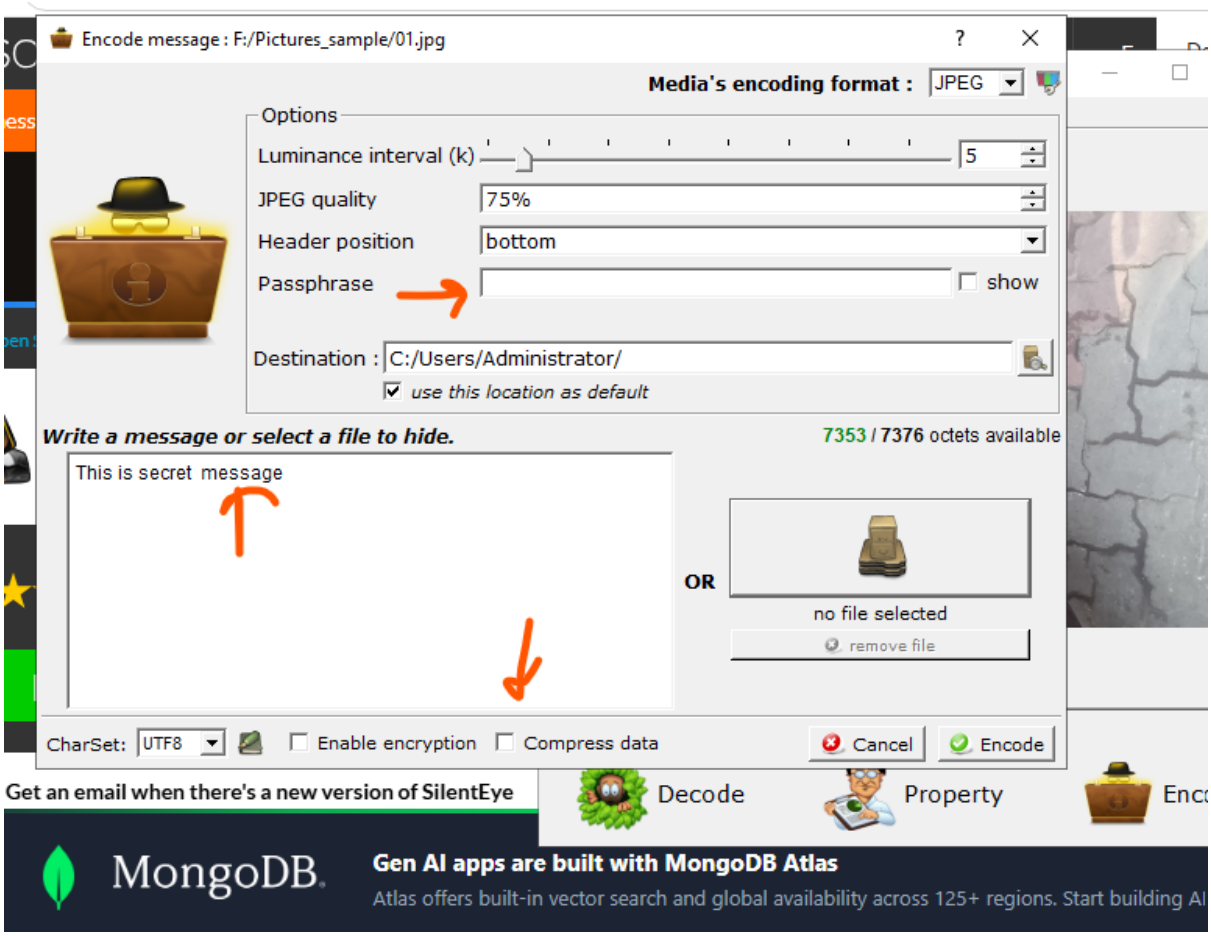
4.8 Navigate to Pictures_sample folder and select the 01.jpg and click on Open



4.9 Click on Encode

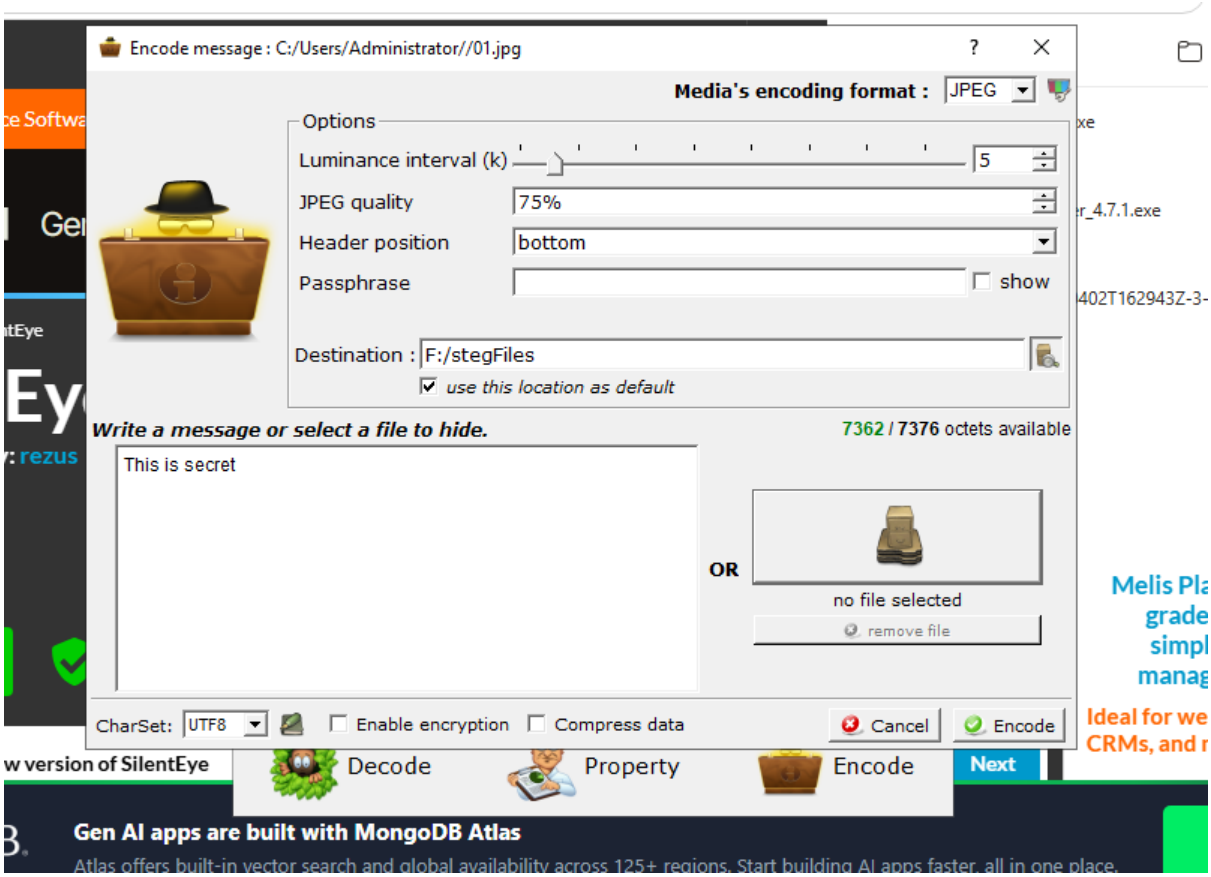


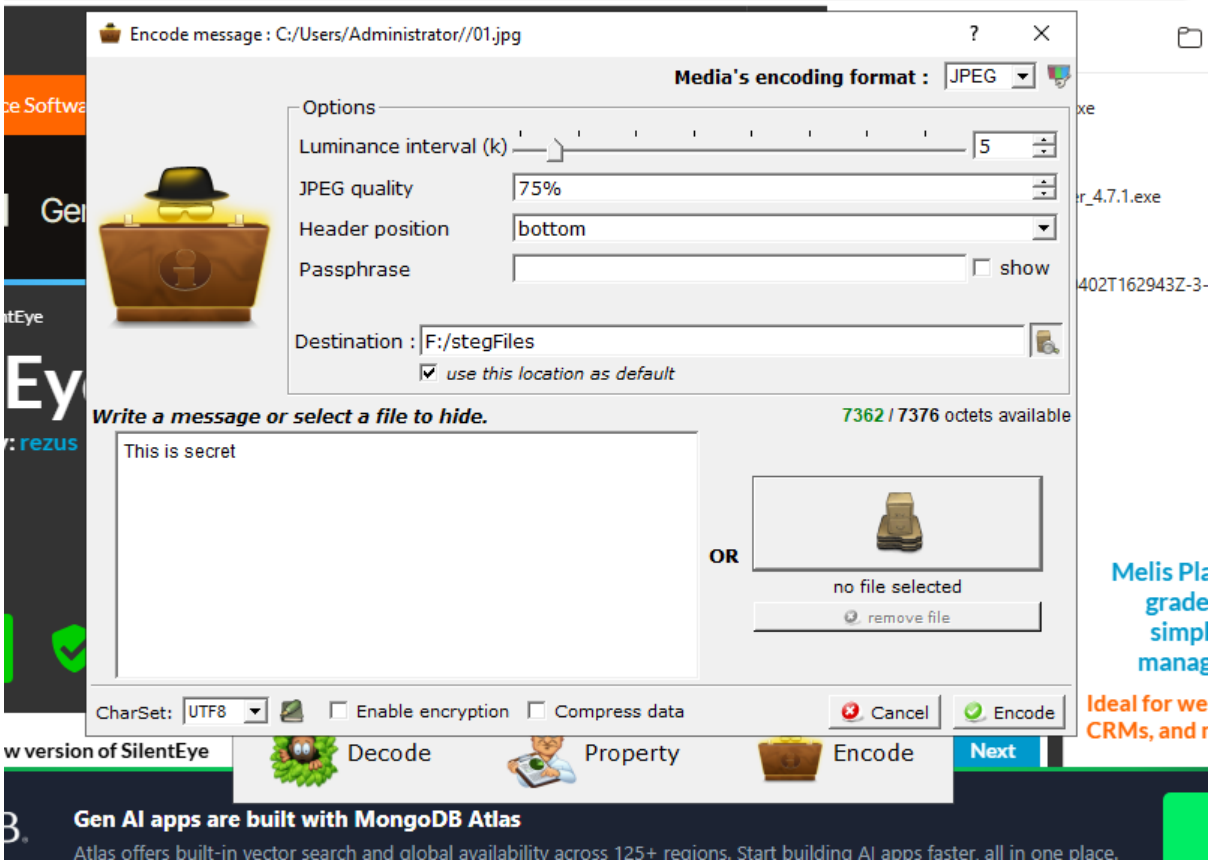
4.10 Remove the Passphrase , uncheck the Compressd at a button , and write the message to be hidden



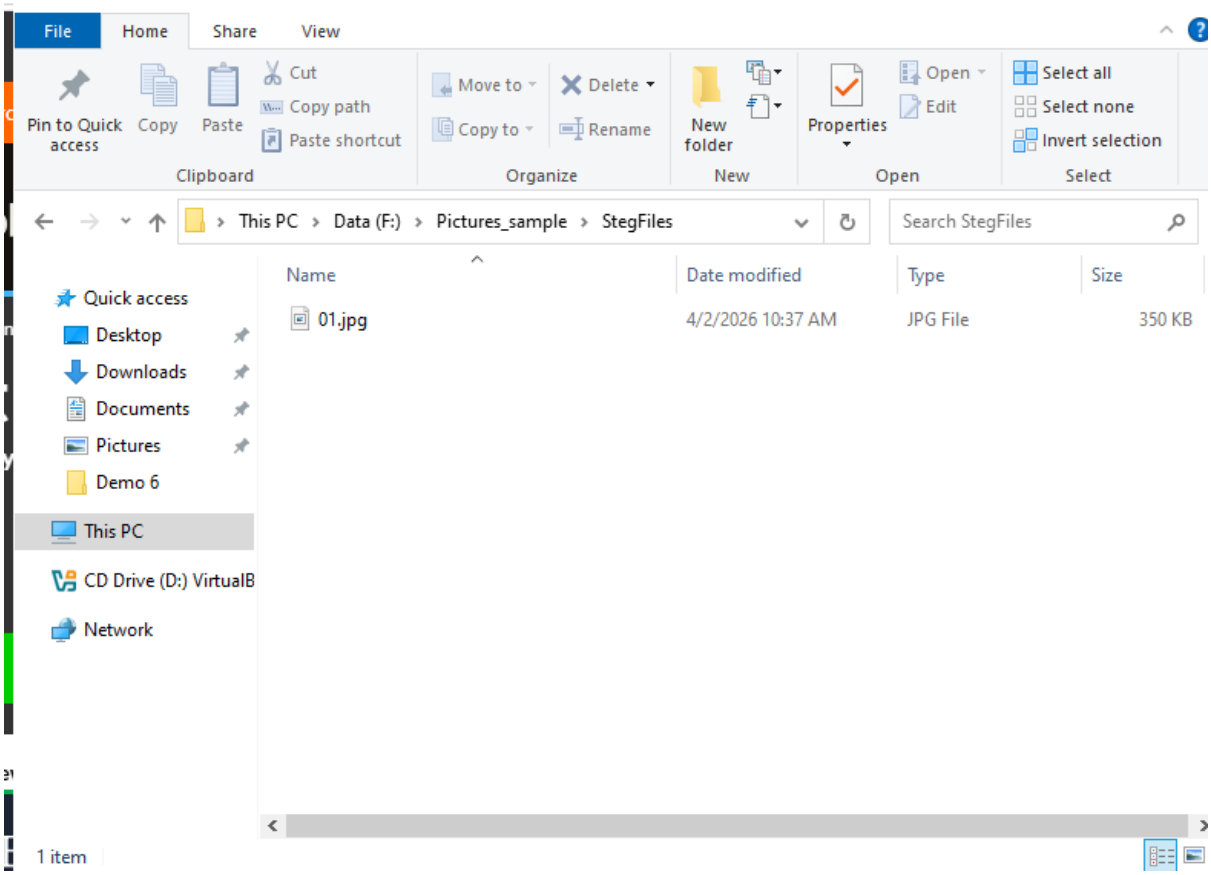
Step 5: Store the image on a 2GB Disk in NTFS

5.1 Change the Destination as Data drive(F), create a folder and give the name as StegFiles, and then click on Encode

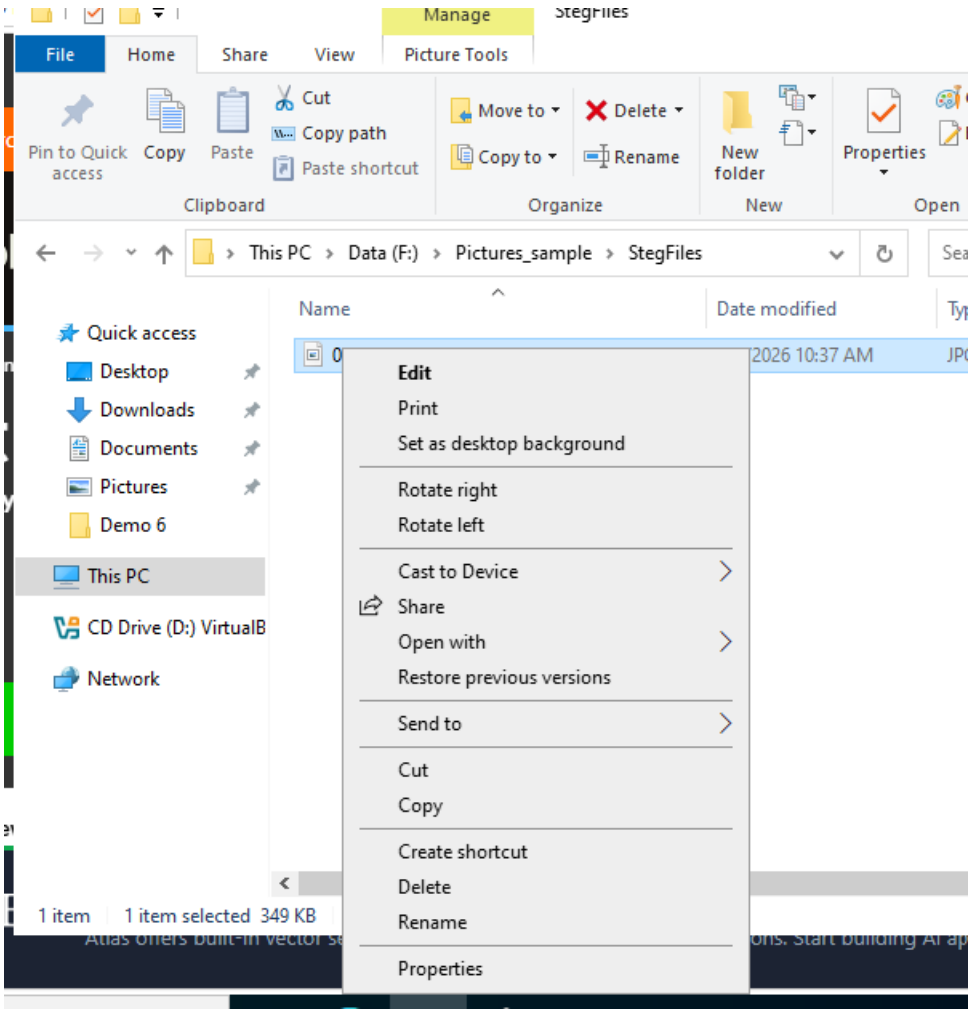

Encode message : C:/Users/Administrator//01.jpg
Media's encoding format : JPEG
Options
Luminance interval (k) 5
JPEG quality 75%
Header position bottom
Passphrase show
Destination : F:/stegFiles
use this location as default
Write a message or select a file to hide.
7362 / 7376 octets available
This is secret
OR
no file selected
remove file
CharSet: UTF8
Enable encryption
Compress data
Cancel
Encode
Decode
Property
Encode
Next
Gen AI apps are built with MongoDB Atlas


Encode message : C:/Users/Administrator//01.jpg
Media's encoding format : JPEG
Options
Luminance interval (k) 5
JPEG quality 75%
Header position bottom
Passphrase show
Destination : F:/stegFiles
use this location as default
Write a message or select a file to hide.
7362 / 7376 octets available
This is secret
OR
no file selected
remove file
CharSet: UTF8
Enable encryption
Compress data
Cancel
Encode
Decode
Property
Encode
Next
Gen AI apps are built with MongoDB Atlas

5.2 Navigate to Data(F): > Pictures_sample > StegFiles. Now, we can see that the file is created successfully

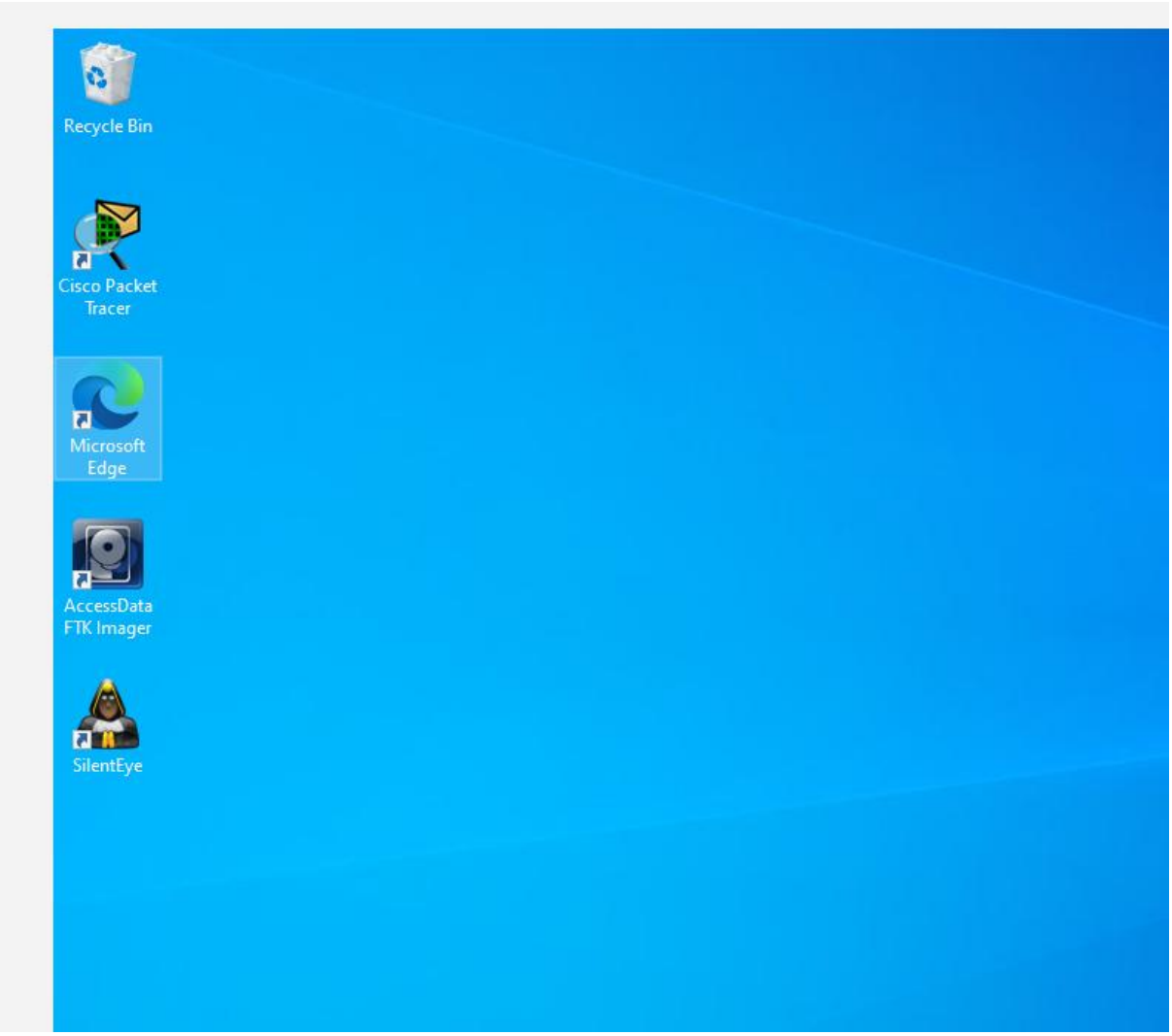


5.3Right-clickonthefileandclickon Delete

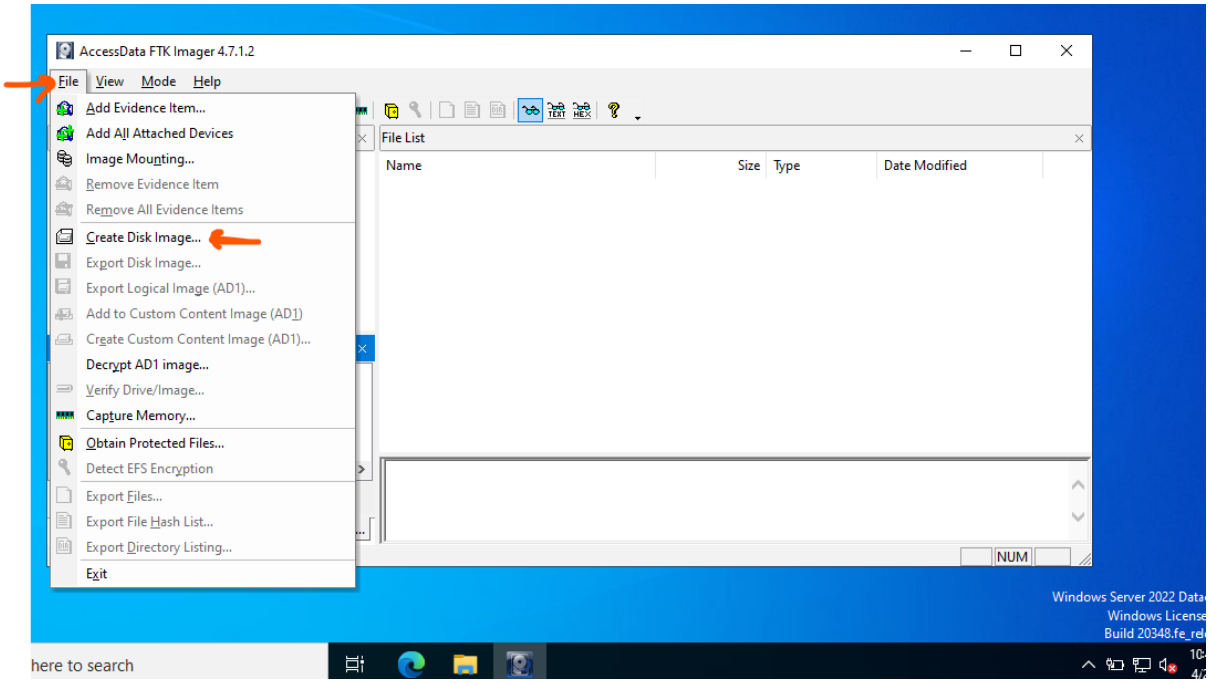


Step6:RecoverthedeletedfileusingFTK

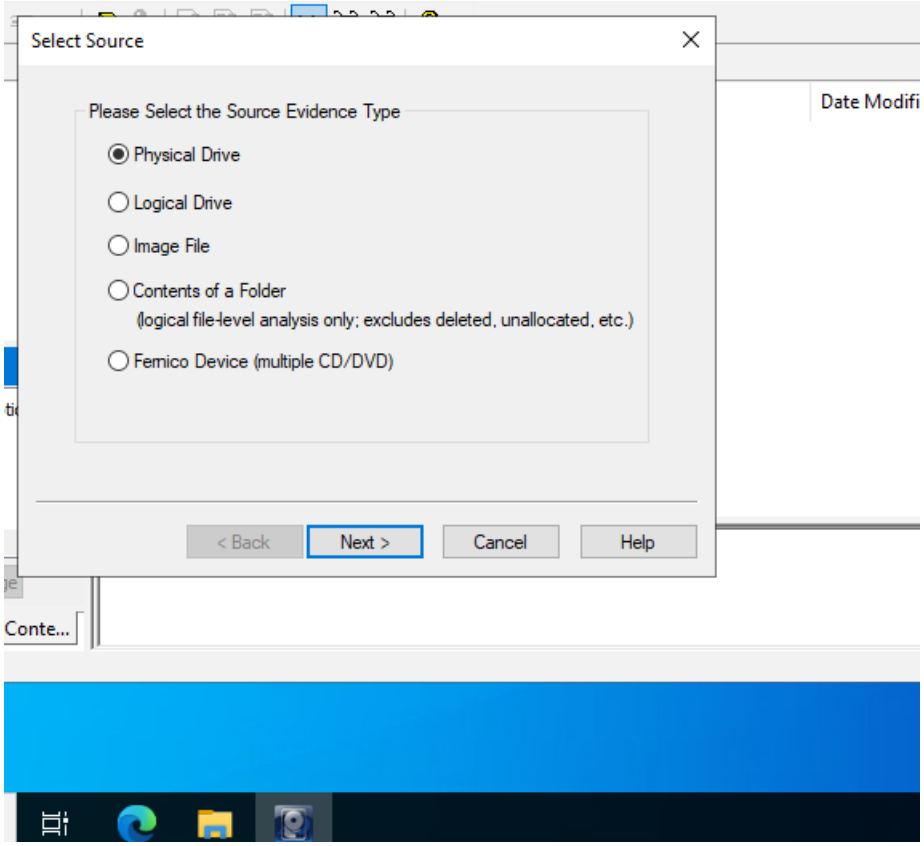
6.1 OpentheAccessDataFTKImager



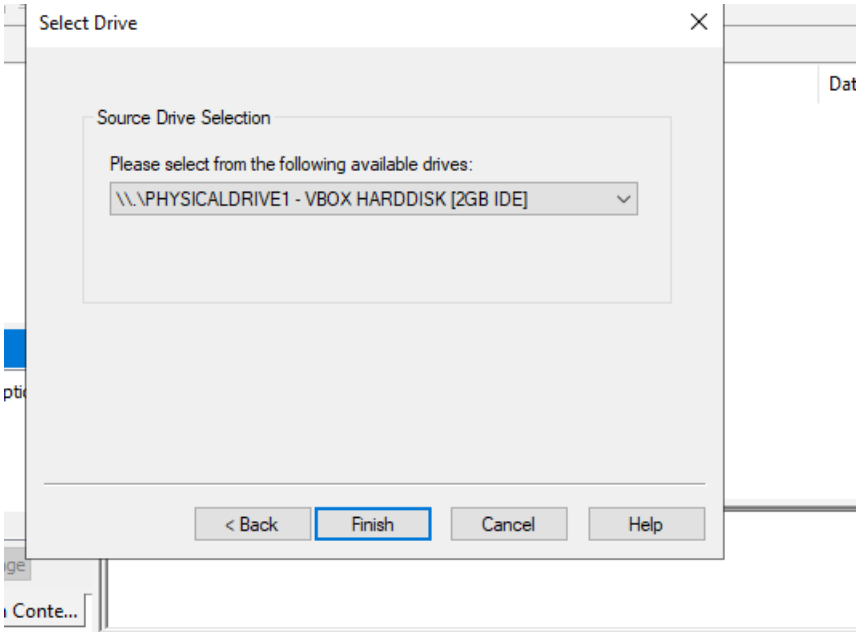
6.2 Click on File and select the Create Disk image..



6.3 Check the radio button of Physical Drive and click Next

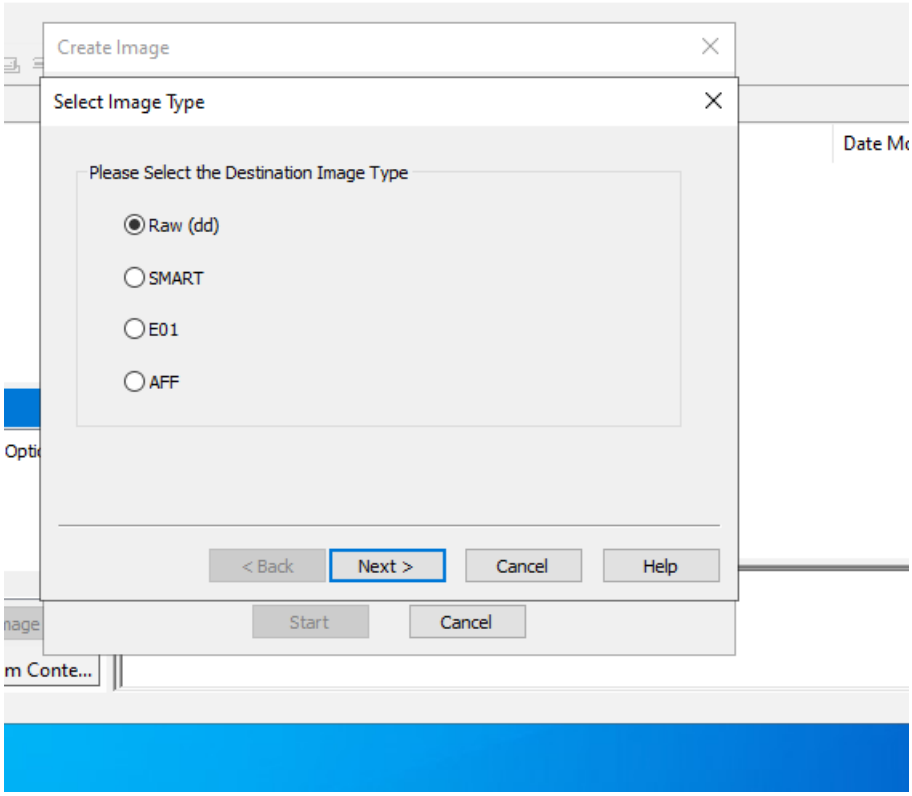


6.4 Choose the 2 GBfile from the drop-down button and click Finish

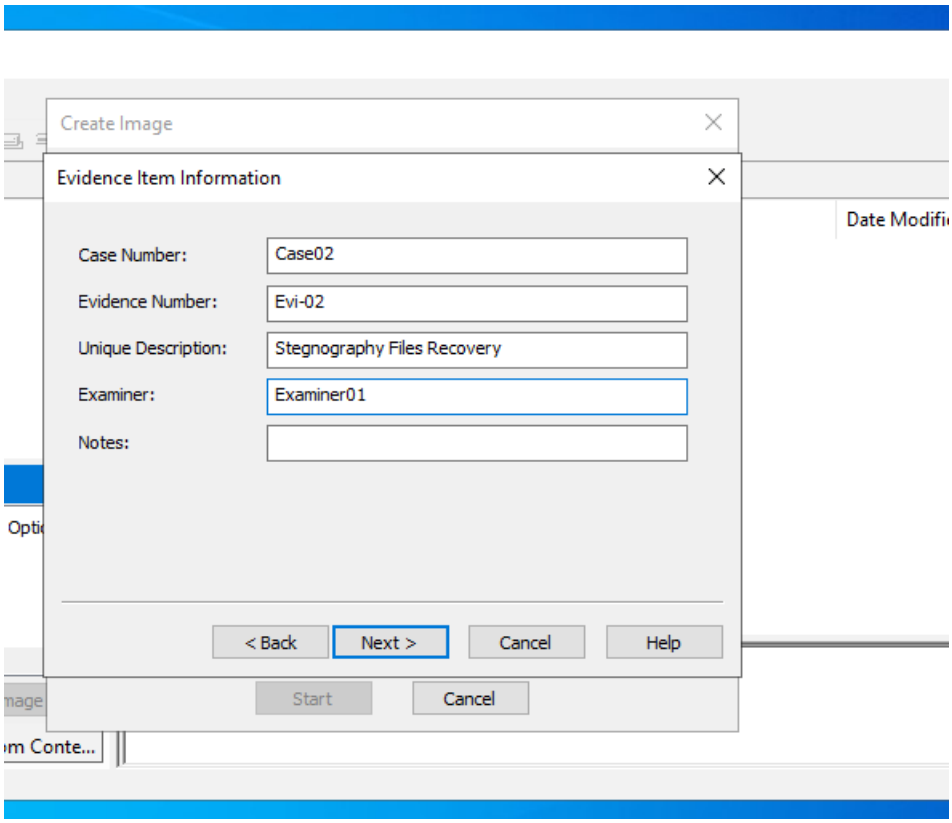


6.5 Click on Add

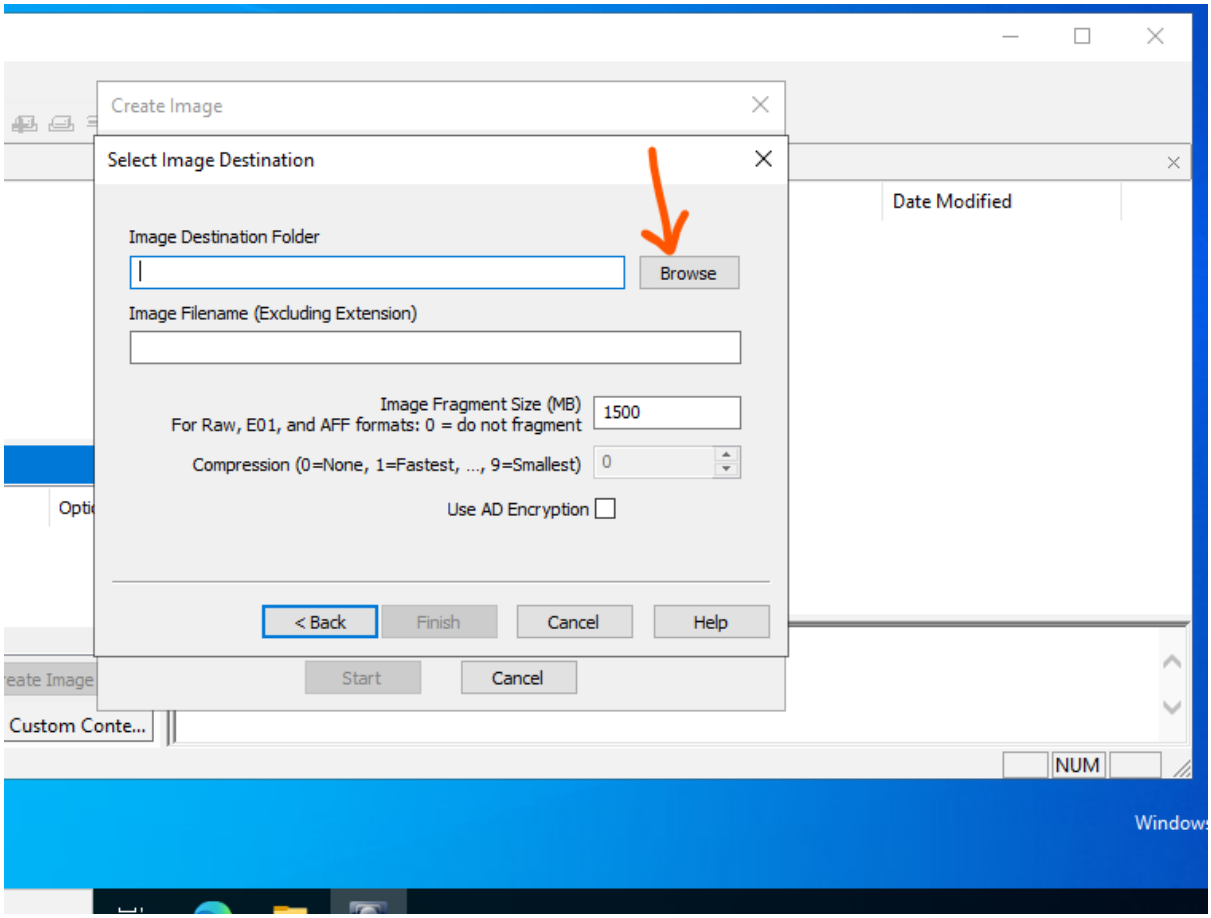
6.6 Check the radio button of Raw and click Next



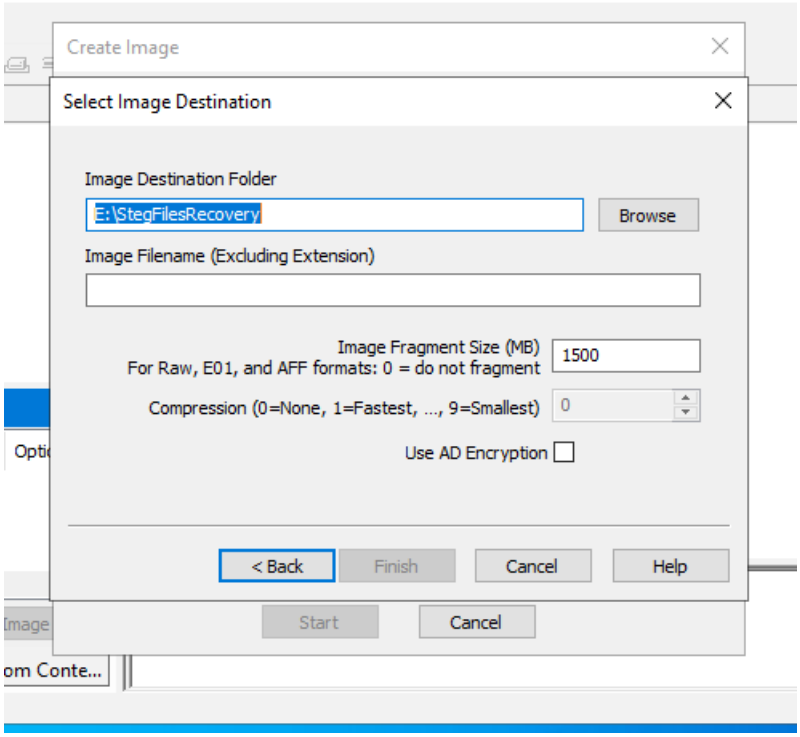
6.7 Enter the evidence information as shown below and click Next



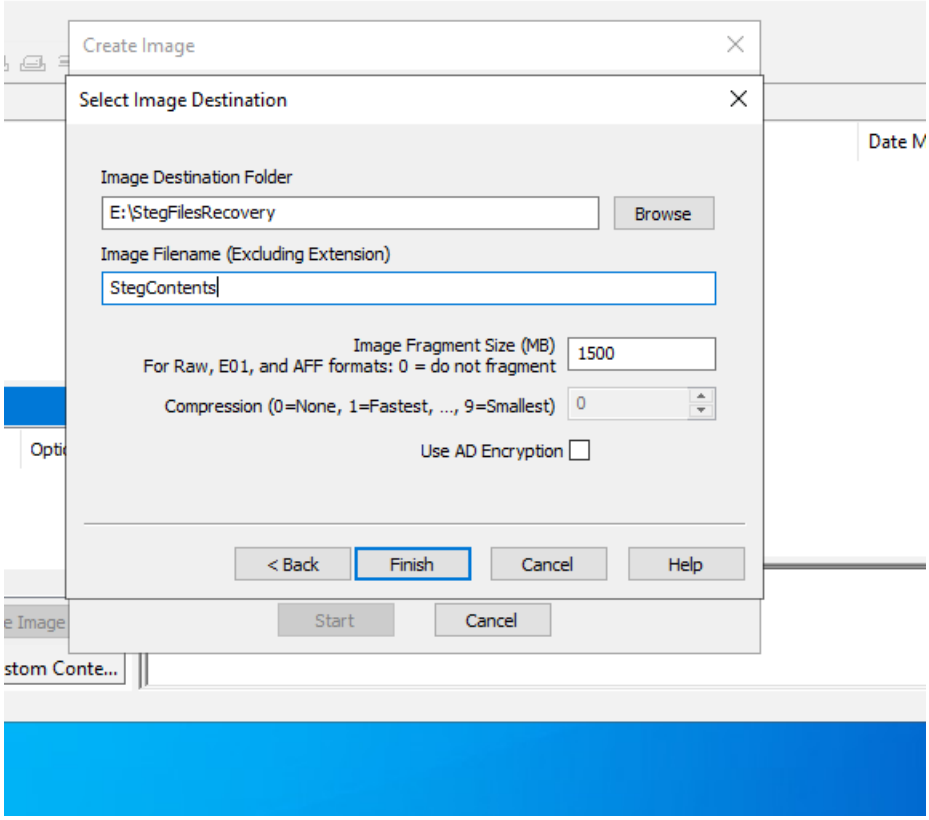
6.8 Click on Browse



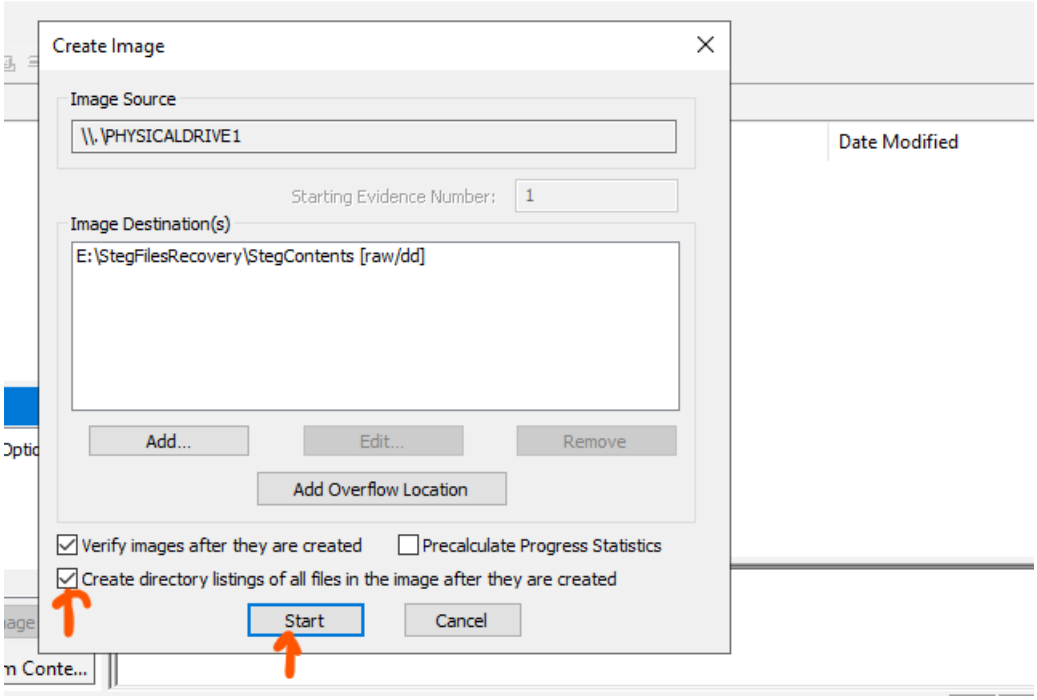
6.9 Create a new folder (StegFilesRecovery) inside the Evidence drive ,select the folder ,and then click on OK

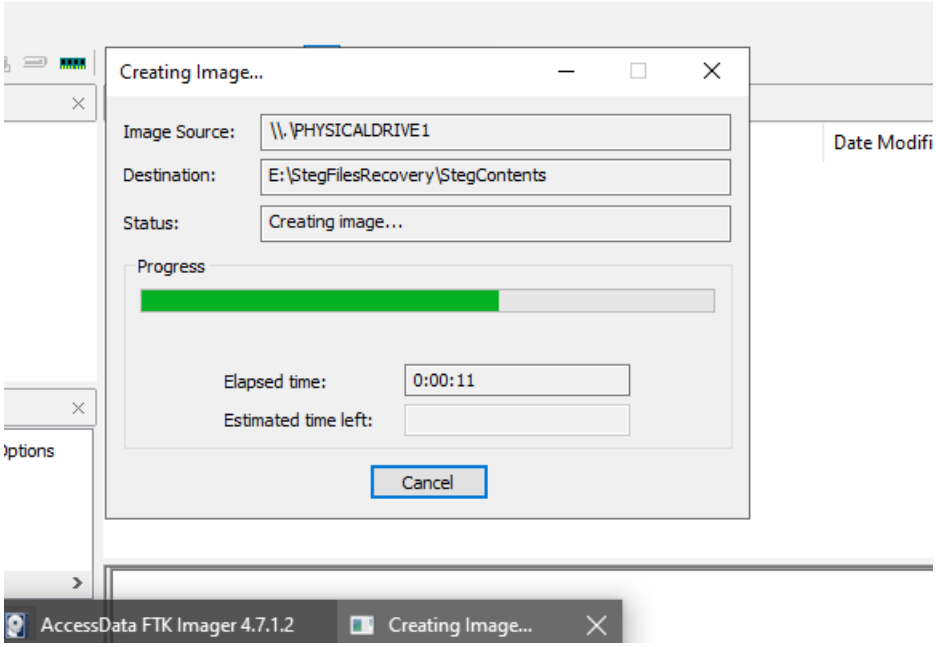


6.10 Enter the Image File name and click on Finish

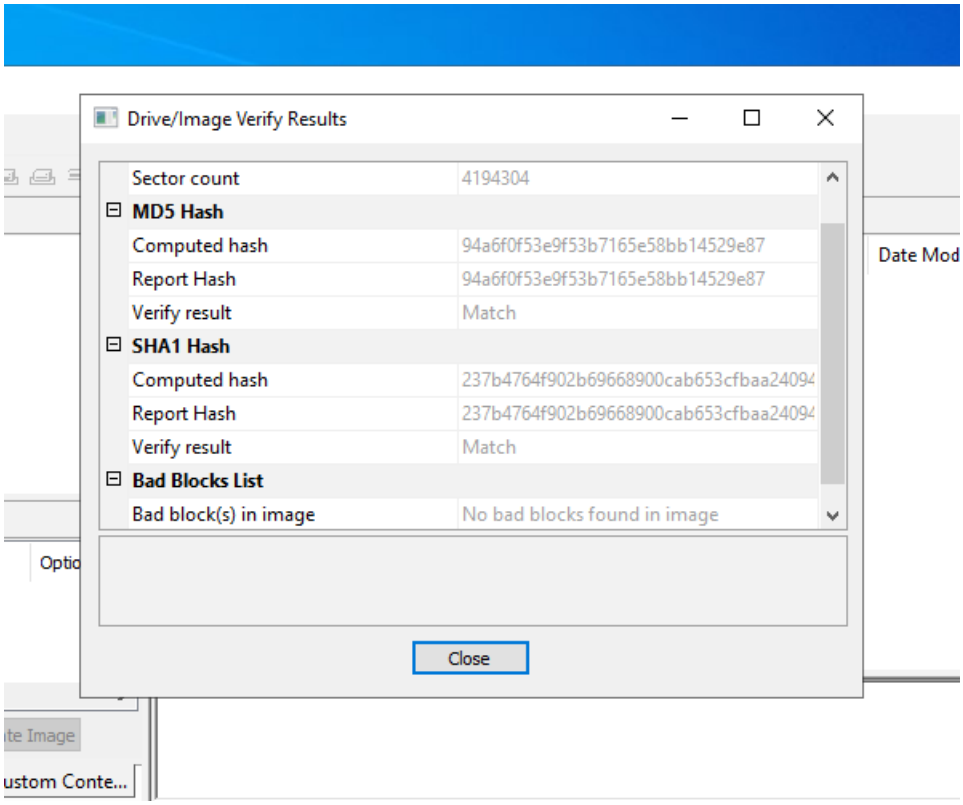


6.11 ChecktheboxCreate directory listings of all files in the image after they are created and click on Start

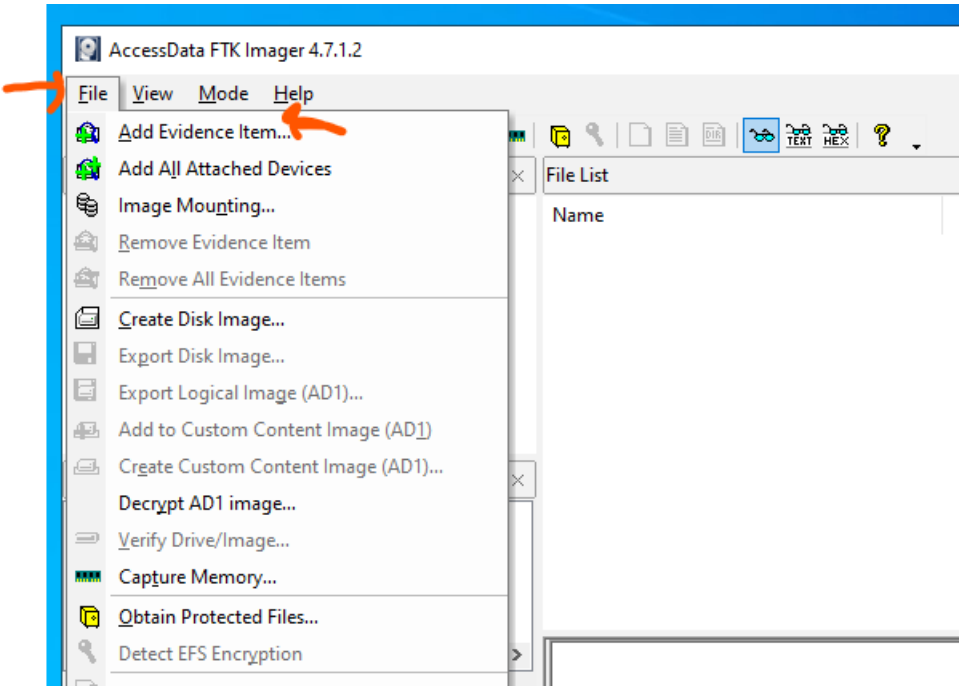




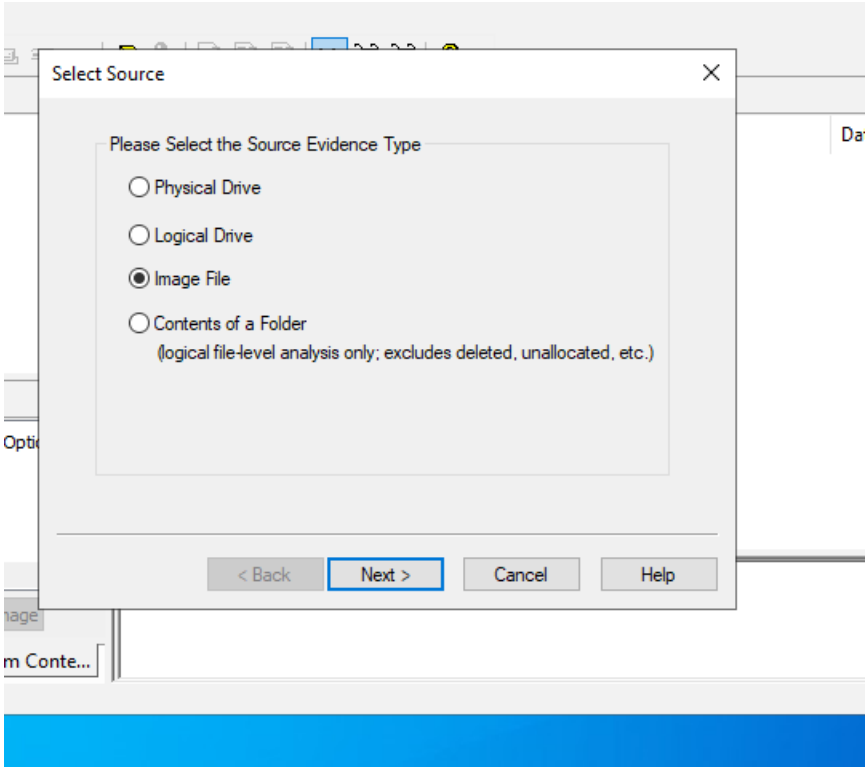
6.12 Click on Close



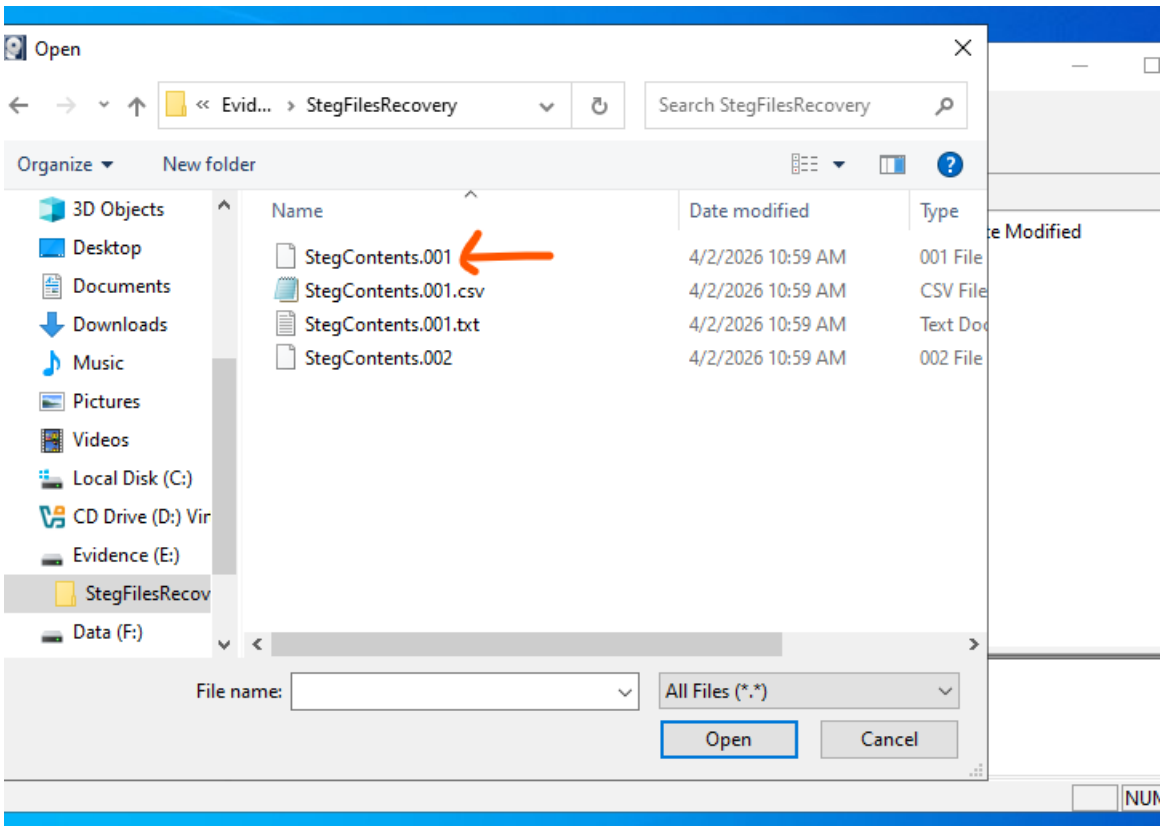
6.13 Click on File and select the Add Evidence Item..



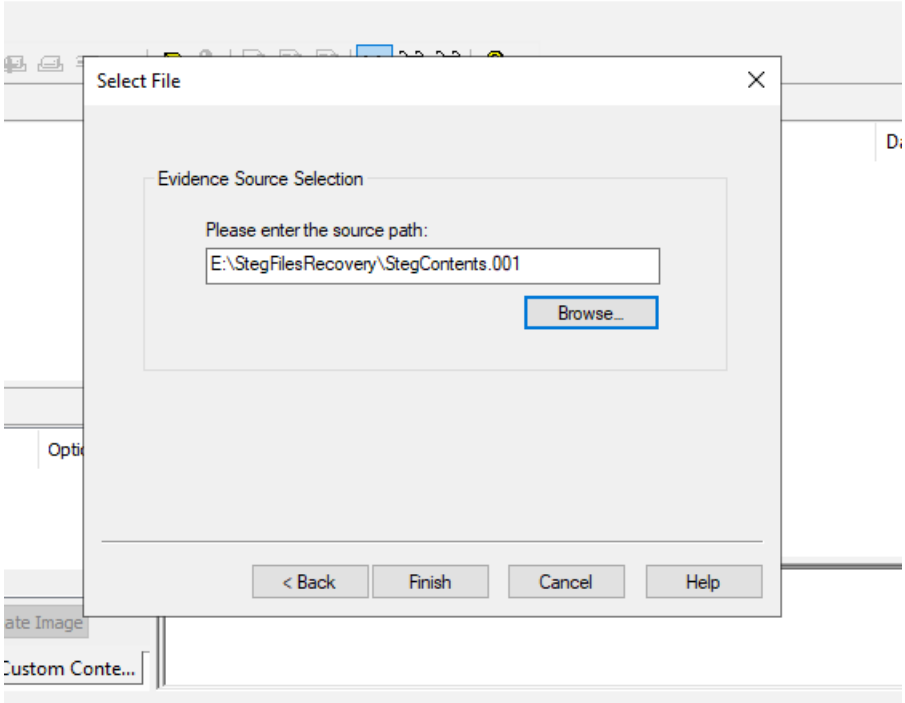
6.14 Check the radio button of Image File and click on Next



6.15 Click on Browse...



6.16 Click on Finish



By following these steps, I have successfully demonstrated the significance of data hiding and its impact on digital forensics. This exercise highlights the use of steganography to conceal data and the role of forensic tools in recovering it, emphasizing the importance of securing data and maintaining forensic readiness in digital investigations